
Impossible Questions

《不可能的问题》

双语对照版 · Bilingual

Scientific American special package

双语对照

October 2026

Bilingual edition

Contents

Finding meaning in the impossible / 在不可能 中寻找意义	1
The questions science may never answer / 科学 或许永远无法回答的问题	6
Will we ever know how life began? / 我们能否 知道生命如何起源?	10
Will we ever achieve immortality? / 人类能否 永生?	30
What if we never find dark matter? / 如果我们 永远找不到暗物质?	46
The hunt for reality's smallest pieces / 追寻实 在最小的碎片	64
What if 'consciousness' isn't real? / 如果「意识」 并不真实?	73
What came before the big bang? / 大爆炸之前 是什么?	80
Why we love unexplainable mysteries / 我们为 何热爱无解之谜	93

Finding meaning in the impossible

在不可能中寻找意义

Finding meaning in the impossible

By Jeanna Bryner · September 15, 2026 · edited by Jeanna Bryner

<https://www.scientificamerican.com/article/finding-meaning>

Some questions may never be answered, but maybe that's not the point

有些问题或许永远得不到答案，但也许那并不是关键所在

There are questions in science that have been studied for so long without anyone turning up answers that they seem to move into the realm of "unsolvable" or even "impossible." This is the realm we explore in Scientific American's October issue.

科学中有一些问题，人们研究了很久却始终找不到答案，它们似乎已进入了“无法解决”甚至“不可能”的领域。这正是《科学美国人》十月号所探索的疆域。

When the editors first discussed this idea, a real concern became immediately apparent: Will our readers be disappointed at the very premise of an unanswerable question? Why dive in if we can't satisfy their, and our, curiosity with a solution?

编辑们最初讨论这一设想时，一个切实的担忧立刻显现出来：读者会不会对“无法回答的问题”这一前提本身感到失望？既然无法以一个解答来满足他们——以及我们自己——的好奇心，为什么还要一头扎进去呢？

But we soon realized that this open-endedness is part of the appeal. Often the most fascinating questions are the ones with no obvious or direct rewards. Unlike figuring out the odds that someone will have a green-eyed baby, the knotty, impossible questions sit in uncharted territory. Scientists must approach them from many different angles, some of which require new tools and experiments never before conducted. Such undirected searching is exactly what fuels serendipity.

但我们很快意识到，这种开放性恰恰是其魅力所在。往往最引人入胜的问题，恰恰是那些没有明显或直接回报的问题。与计算某人生下绿眼睛婴儿的概率不同，这些棘手的、不可能的问题坐落在未知的疆域。科学家必须以许多不同的角度切入它们，其中一些需要从从未有过的新工具和新实验。正是这种无明确指向的探索，滋养着偶然的发现。

We probably wouldn't have the blockbuster GLP-1 weight-loss drugs now reshaping medicine, for instance, if someone didn't ask how Gila monsters cheat starvation—the lizards' blood sugar never collapses even though they sometimes go months without eating. There was no obvious payoff to finding the answer and no pharmacology at the time that could explain the anomaly.

举个例子：若不是有人追问吉拉毒蜥如何逃避饥饿——这些蜥蜴有时数月不进食，血糖却从不崩溃——我们大概不会拥有如今正在重塑医学的重磅 GLP-1 减肥药。当时寻找答案并无明显回报，也没有药理学能解释这一异

常。

Alchemists searched for centuries for a way to turn common metals into precious ones such as gold—an impossible challenge when relying on a philosopher's stone. Along the way, they established the field of chemistry and discovered the process of distillation that is ubiquitous today in water purification, pharmaceutical production and even fragrance making. (Physicists at the Large Hadron Collider at CERN near Geneva accidentally did turn lead into gold recently, at least for a fraction of a second.)

炼金术士们数百年间寻找把普通金属变成黄金等贵金属的方法——在依赖哲人石的前提下，这是一项不可能的挑战。一路走来，他们奠定了化学这门学科，并发现了蒸馏工艺；如今蒸馏在水净化、制药乃至香水制造中无处不在。（日内瓦附近欧洲核子研究中心大型强子对撞机的物理学家们，最近也意外地把铅变成了金——至少持续了几分之一秒。）

For more than 2,000 years mathematicians chased a problem called squaring the circle, in which they tried to craft a square with the same area as a circle, using only a compass and a straightedge. Proof that it was impossible came with the discovery of transcendental numbers, which led to theorems that opened entirely new branches of mathematics.

两千多年来，数学家们追逐一个叫作“化圆为方”的问题：仅用圆规和直尺，做出一个与给定圆面积相等的正方形。证明其不可能，伴随着超越数的发现，而这些发现又引出了开辟数学全新分支的定理。

In the special report, philosopher Tim Bayne asks whether consciousness is real, a question that epitomizes the word “intractable”: the very act of studying it requires the

faculty it's trying to explain. But plenty of scientists have dedicated their lives to finding an answer. Along the way, that work has refined the tools we use to keep people safely unconscious under anesthesia. Other researchers turned to LSD and psilocybin to probe consciousness at its edges and found, serendipitously, that these substances carry real therapeutic effects for depression, post-traumatic stress disorder, and other resistant conditions.

在本期特别报道中，哲学家蒂姆·贝恩追问意识是否真实——这个问题堪称“棘手”一词的典范：研究它本身就需要诉诸它所试图解释的那种能力。但许多科学家毕生致力于寻找答案。一路上，这些工作改进了我们在麻醉下安全使人失去意识的工具。另一些研究者转向 LSD 与裸盖菇素，在意识的边缘加以探查，却意外发现这些物质对抑郁症、创伤后应激障碍及其他难治病症确有治疗效果。

The same pattern can be seen in the decades physicists have spent looking for the constituents of dark matter. The still unanswered question has led to ever more sensitive detector technology and a deeper understanding of what dark matter isn't.

同样的模式，也见于物理学家们数十年来寻找暗物质组成的努力。这个仍未解答的问题，催生了越来越灵敏的探测器技术，也让我们对“暗物质不是什么”有了更深的理解。

Although these "discoveries along the way" are meaningful, they might not even be the primary reason to pursue impossible questions. As physicist and Scientific American senior reporter Joseph Howlett writes of the search for dark matter, "Maybe the search for absolute truth—

not just a reliably predictive picture of the world but a meaningful one—was always too ambitious.”

尽管这些“沿途发现”意义重大，它们甚至可能并非追逐不可能问题的首要理由。正如物理学家、本刊资深记者约瑟夫·豪利特在写到搜寻暗物质时所说：“或许，对绝对真理的追寻——不只是一幅可可靠预测的世界图景，而是一幅有意义的图景——从来都过于野心勃勃。”

Perhaps the journey we take in our search for meaning is the goal. Humans are wired for curiosity and social connection. At the end of the day, impossible questions might satisfy both because they require us to ask why, how and what. All of science builds on the discoveries that came before and the people who made them. The intractable part could be the very thing that gives our lives meaning.

也许，我们在追寻意义的旅途中所走的路，本身就是目的。人类天生渴望好奇与社会联结。归根结底，不可能的问题或许能同时满足这两者，因为它们迫使我们去追问为什么、如何、是什么。全部科学都建立在前人的发现以及做出这些发现的人之上。那棘手的部分，或许正是赋予我们生命以意义的东西。

The questions science may never answer

科学或许永远无法回答的问题

The questions science may never answer

By Sarah Scoles · September 15, 2026 · edited by Clara Moskowitz

<https://www.scientificamerican.com/article/the-questions-s>

Will some of the biggest mysteries in science remain forever unsolved?

科学中一些最大的谜题，会永远无解吗？

Science, at its best, is a striving toward truth. It is not, although we often think of it as such, the search for absolute truth. It's more a process of systematically gathering and interpreting data about the universe to understand its past and present and predict its future. What results is the best approximation we can access at any given time with the information, methods and instruments we have.

科学在其最佳状态，是对真理的奋力趋近。它并非——尽管我们常这样想——对绝对真理的搜寻。它更像是一个系统收集并诠释关于宇宙的数据的过程，以理解其过去与现在，并预测其未来。所得的结果，是我们在任何给定时刻，凭当时拥有的信息、方法与仪器所能触及的最佳近似。

And what counts as the best approximation of truth often changes. Isaac Newton began to devise his apple-falling theory of gravity in 1666. It worked well for everyone's purposes in the 17th century. Then about 250 years later Albert Einstein stepped in with a better approximation: the general theory of relativity. Today some scientists are working to find ways to unite relativity with quantum mechanics, seeking a new approximation of the "truth" of gravity.

而何谓真理的最佳近似，也常常会改变。艾萨克·牛顿在 1666 年开始构思他那套苹果落地的引力理论。它很好地满足了 17 世纪所有人的用途。大约 250 年后，阿尔伯特·爱因斯坦带着更好的近似登场：广义相对论。今天，一些科学家正设法把相对论与量子力学统一起来，寻求对引力之“真理”的新近似。

Not all questions will find answers. Some will likely remain beyond scientific grasp, even with methodological and interdisciplinary advances.

并非所有问题都会找到答案。即便有方法论与跨学科的推进，有些问题也很可能永远超出科学的把握。

But are there some questions science can never fully answer? There are certainly questions scientists can't answer yet. There are also queries scientists once thought impossible to resolve that weren't. In 1835, for instance, philosopher Auguste Comte asserted that humans could never know what stars are made of. Then came spectroscopy, which allowed astronomers to parse stellar light for fingerprints of different chemicals.

但是否存在科学永远无法完全回答的问题？当然有科学家们目前还答不上来的问题。也有科学家一度以为不可能解决、后来却解决了的疑问。例如 1835 年，哲学家奥古斯特·孔德断言人类永远无法知道恒星由什么构成。随后光谱学出现了，天文学家得以解析星光中不同化学物质的指纹。

Most things that seem forever unknowable eventually fall out of that porous basket. New ways of thinking or measuring expand the boundaries of science. This idea is the thesis of scholar Alexander Krauss's most recent book, *The Engine of Scientific Discovery*. Until radio antennas existed, for instance, astronomers didn't know the universe emitted radio waves, so they didn't wonder what powered the gigantic jets shooting at nearly light speed from the centers of galaxies. Before microscopes, biologists didn't know about animalcules, as microbes were first called, so they didn't know they might someday study how the microbiome impacts mental health. "When scientists invent novel instruments and new computational and experimental techniques, they effectively expand the sensory and cognitive reach of science itself," Krauss says.

大多数看似永远不可知的事物，最终都会从那只多孔的篮子里掉出来。新的思维方式或测量手段扩展着科学的边界。这正是学者亚历山大·克劳斯近著《科学发现的引擎》的论题。例如，在射电天线出现之前，天文学家并不知道宇宙会发出射电波，因此也就不会去想是什么驱动了从星系中心几乎以光速喷出的巨大喷流。在显微镜出现之前，生物学家不知道“微动物”——微生物最初的称呼——因此也不知道有朝一日他们可能研究微生物组如何影响心理健康。“当科学家发明新仪器以及新的计算与实验技术时，他们实际上扩展了科学本身的感官与认知触角，”克劳斯说。

Still, not all questions will find answers. Some will likely remain beyond scientific grasp, even with methodological and interdisciplinary advances. Some will be so beyond our imagination that they will remain unknown unknowns forever. That's frustrating: science aims to understand the universe by observing, experimenting, interpreting, iterating. But that enterprise hits walls when there is nothing to observe or experiment on, when something lies past a fundamental barrier, or when observing or experimenting changes what is being observed or experimented on.

尽管如此，并非所有问题都会找到答案。即便有方法论与跨学科的推进，有些问题也很可能永远超出科学的把握。有些问题会远超我们的想象，以至于永远是未知的未知。这令人沮丧：科学旨在通过观察、实验、诠释、迭代来理解宇宙。但当无可观察、无可实验，当事物落在某个根本屏障之外，或当观察与实验本身改变了被观察、被实验的对象时，这项事业就会撞上墙壁。

So how do we make sense of, and make peace with, necessary gaps in our knowledge? Perhaps by accepting that science isn't about finding the truth, the whole truth, and nothing but the truth. We'll just have a set of evolving best shots and, ideally, shots that keep leading—like a non-Brownian walk—to the next destination.

那么，我们如何理解、并与知识中这些必然的空白和解？或许要接受：科学并非要找出全部真相、唯有真相。我们只会拥有一套不断演化的最佳尝试——理想情况下，这些尝试会像非布朗运动那样，持续导向下一站。

Will we ever know how life began?

我们能否知道生命如何起源？

Will we ever know how life began?

By Sarah Scoles · September 15, 2026 · edited by Clara Moskowitz

<https://www.scientificamerican.com/article/will-we-ever-know-how-life-began/>

Scientists disagree on how biology got started and on whether we can ever know for sure

科学家们对生物学如何起步、以及我们是否终能确知，意见不一

In a laboratory at McMaster University in Ontario, a device that seems almost alive glows purple as it churns through its work. Atop a boxy base that resembles a small kitchen island, a windowed hemispherical chamber sits. Cords and instruments poke from it like flower petals. Gas billows from its bottom. The setup's purpose is hard to fathom if you don't know what it is and even perhaps if you do. Inside that little chamber is a whole world. Its name is, appropriately, the Planet Simulator.

在安大略省麦克马斯特大学的一间实验室里，一台几乎像有生命的装置发着紫光，不停运转。箱状底座形似一座小型厨房岛台，其上安着一间带窗的半球形舱室。线缆与仪器像花瓣一样从中探出。气体从底部涌出。若你

不知其为何物——即便知道——也很难揣度这套装置的用途。那小小舱室里，是一整个世界。它的名字恰如其分：行星模拟器。

Maikel Rheinstädter, a professor in McMaster's physics and astronomy department, and his team can program the simulator to mimic many different environments. They can swing it through temperatures from below freezing to past boiling, change the ingredients of the tiny atmosphere, create rain and then dry it up, and shine light ranging from ultraviolet to infrared, as different stars would. In the chamber's center, a drop of water on a chip contains the chemistry of whatever earthly era or exoplanetary site the group is trying to re-create.

麦克马斯特大学物理与天文学系教授迈克尔·莱因施泰特及其团队，可以编程让模拟器模仿许多不同的环境。他们能让温度从冰点以下一路摆到沸点以上，改变微型大气的成分，制造雨水再把它烘干，并投射从紫外到红外的光线，宛如不同恒星所为。在舱室中央，芯片上的一滴水包含着团队试图重现的任何地球时代或系外行星地点的化学条件。

The scientists are watching to see what might grow in that little bit of liquid—to determine which ingredients and conditions can produce life's building blocks, get them to stick together, and maybe someday bring something to life. If it can do that, the Planet Simulator could help answer one of the biggest questions of all: How did life start on Earth?

科学家们注视着那一小滴液体里可能长出什么——以判断哪些原料与条件能产生生命的构件，让它们粘在一起，或许有朝一日造出生命。若真能做到，行星模拟器或许有助于回答最大的问题之一：地球上的生命是如何开始

的？

But whether any scientists, even with whole small worlds under their command, can definitively find the answer to that question is another matter. Can we truly know the origins of our planet's life, or will we at best be able to find only plausible possibilities? This larger query remains an open question. After all, the natural experiment that first produced life is not reproducible, no matter how fancy labs become.

但即便科学家手握整座小型世界，他们是否真能明确找到那个问题的答案，则是另一回事。我们能否真正知晓这颗地球上生命的起源，还是充其量只能找到若干貌似合理的可能？这一更大的疑问仍悬而未决。毕竟，最初产生生命的那场自然实验无法重现，无论实验室多么精巧。

Some questions are human classics, asked for millennia and more. Why are we here? Are we alone? What happens when we die? Where did we come from? In some way, all those quandaries are behind the one Rheinstädter is pursuing: What is life, and how did it—we—get here in the first place?

有些问题是人类的经典，被追问了数千年乃至更久。我们为何在此？我们是否孤独？死后会发生什么？我们从何处来？在某种意义上，莱因施泰特所追问的那个问题背后，正是所有这些困惑：生命是什么，它——我们——最初又是如何来到这里的？

In some traditions, the answer is spiritual. Creation stories are, after all, fundamental to most religions. There is even a kind of scientific creation story, a theory of life called spontaneous generation, most prominently proposed by Aristotle more than 2,000 years ago. Sponta-

neous generation is the idea that life can poof! arise from not-life: carcasses or trash beget maggots; scallops and clams grow in sandy spots. That thinking dominated for millennia, until Louis Pasteur sterilized some broth in 1859 and placed it in a glass container. When he cut the liquid off from outside contamination, nothing grew. But when the outside world was allowed in, the broth became quickly cloudy—teeming with microbes. Pasteur’s experiment implied that being alive is hereditary—life, not not-life, begets life.

在某些传统中，答案是灵性的。创世故事毕竟是多数宗教的根基。甚至还有一种科学式的创世故事：亚里士多德两千多年前最突出地提出的生命理论——自然发生说。自然发生说认为，生命可以从非生命中“噗”地一下冒出来：尸体或垃圾生出蛆虫；扇贝与蛤蜊在沙地里长出。这种想法主导了数千年，直到路易·巴斯德于 1859 年把一些肉汤灭菌，放入玻璃容器。当他把液体与外界污染隔绝时，什么也不长。但当外界被允许进入，肉汤很快变得浑浊——微生物蜂拥而至。巴斯德的实验暗示：活着是遗传的——生命，而非非生命，产生生命。

But if that were the case, something still had to be the first alive thing. In that earliest instance, then, the idea of spontaneous generation isn’t actually so far off: at some point, the chemical and physical did have to unite in such a way that they became biological. “When you learn about biology in school, it’s taught as this completely different discipline than something like chemistry or physics,” says Harrison B. Smith, a complex systems and astrobiology researcher at Tokyo’s Earth-Life Science Institute. “But we have to remember that biology emerged still subject to the same constraints of chemistry and physics.”

可若果真如此，仍须有某个东西是第一个活物。在那最

初的一刻，自然发生说其实并不那么离谱：在某个节点，化学与物理必须以某种方式结合，变成生物学。“在学校学生物时，它被教成一门与化学或物理完全不同的学科，”东京地球生命科学研究所复杂系统与天体生物学者哈里森·B·史密斯说。“但我们必须记住，生物学的出现，仍然受制于化学与物理的相同约束。”

Just when it emerged, and how, is tricky to pin down. Despite decades of increasingly complex research, a 2020 review noted that a then 45-year-old proclamation was still true. "It must be admitted from the beginning that we do not know how life began," chemists Stanley Miller and Leslie Orgel wrote in their classic 1974 book *The Origins of Life on the Earth*.

它究竟何时出现、如何出现，很难钉死。尽管数十年研究日益复杂，2020 年的一篇综述仍指出，四十五年前的那句断言依旧成立。“必须从一开始就承认，我们不知道生命如何开始，”化学家斯坦利·米勒与莱斯利·奥格尔在其 1974 年经典著作《地球上生命的起源》中写道。

Miller had conducted an experiment in the 1950s showing that if he simulated the chemical conditions of the early Earth—in a proto-version of the Planetary Simulator—and hit the chemicals with electric sparks, organic molecules started to form. "It is generally believed that a variety of processes led to the formation of simple organic compounds on the primitive earth," the quote continues. "These compounds combined together to give more and more complex structures until one was formed that could be called living." But, the authors added, "No one should be satisfied with an explanation as general as this."

米勒在 1950 年代做过一项实验：若他模拟早期地球的化学条件——行星模拟器的一个原型版本——并用电火花

轰击这些化学物质，有机分子便开始形成。“一般认为，多种过程导致了原始地球上简单有机化合物的形成，”引文继续写道。“这些化合物结合起来，给出越来越复杂的结构，直到形成一个可称为生命的结构。”但作者又补充：“任何人都不应满足于如此笼统的解释。”

And scientists aren't.

科学家们也并不满足。

The possibly never-ending quest toward an explanation is the purview of a scientific field called Origins of Life, or OoL (a word that seems like it would be a good name for the first-ever organism). Little in this field is known for certain. There's general agreement on relatively few points and much contention and competition about others.

通向解释的这条或许永无止境的求索，属于一个叫作生命起源 (Origins of Life, 简称 OoL——听起来像是第一个有机体的好名字) 的科学领域。这个领域几乎没有什么确知。相对少数几点上存在大体共识，其余则充满争论与竞争。

As far as consensus goes: Earth's oldest fossils date back to at least 3.4 billion years ago, with evidence that life may have arisen more than half a billion years before that. Scientists also generally agree that life didn't come into being in single flash but in an extended sequence. Over time chemical reactions produced early versions of metabolism, the ability for molecules to pass down information, and the formation of compartments that would become the boundaries of cells.

就共识而言：地球最古老的化石至少可追溯到 34 亿年前，并有证据表明生命可能在那之前半亿多年就已出现。

科学家们也大体同意，生命并非在一次闪光中突然出现，而是经历了一段漫长的序列。随着时间推移，化学反应产生了早期形式的代谢、分子传递信息的能力，以及后来成为细胞膜边界的隔室。

That's about it, though, in terms of what's known. For the rest, scientists can't even agree on the right way to approach open questions. Some researchers take a top-down tack, looking at what modern organisms have in common, such as cell-based structure, DNA and RNA, and specific energy molecules. Those commonalities, the thinking goes, probably stem from properties fundamental to the earliest life.

然而，就“已知”而言，大概也就这些。至于其余，科学家们甚至无法就接近开放问题的正确方式达成一致。一些研究者采取自上而下的路径，考察现代生物的共同点，例如基于细胞的结构、DNA 与 RNA，以及特定的能量分子。按此思路，这些共性很可能源于最早生命的根本属性。

Other scientists go bottom-up, mimicking in a lab what Earth's environment might have been like billions of years ago to try to produce the complex chemistry that became biology, as Miller and Rheinstädter have done. In this approach, scientists don't assume that modern life much resembles early life. Rather modern biochemistry may be largely the result of evolution.

另一些科学家则自下而上，在实验室中模仿数十亿年前地球环境可能的样子，试图产生后来成为生物学的复杂化学——米勒与莱因施泰特所做的正是如此。在这种方法中，科学家并不假定现代生命与早期生命十分相似；相反，现代生物化学可能在很大程度上是演化的结果。

David Catling of the University of Washington is a scientist in the latter group, having come to OoL research through his studies of planets. That started early: In an undergraduate astronomy course, Catling had to come up with a telescope project. "Other people were looking at very faint stars, which you can't even see in the viewfinder," he says. "And I thought, 'Well, I'll do Jupiter. You can't miss that.'" After measuring methane above Jupiter's clouds, he became interested in planets' atmospheres and eventually the climate of early Earth.

华盛顿大学的戴维·卡特林属于后一阵营；他是经由行星研究进入生命起源领域的。那开始得很早：在一门本科天文学课上，卡特林必须想出一个望远镜课题。“别人在看非常暗淡的恒星，取景器里甚至看不见，”他说。“我想：‘好吧，我看木星。那可错过不了。’”在测量了木星云层上方的甲烷之后，他对行星大气产生了兴趣，最终转向早期地球气候。

At first, Catling wasn't sure he had much to contribute to OoL work. "I'm not a biochemist. I'm not an organic-synthesis chemist," he says. But, he learned, chemists needed people like him to do exactly what he was interested in: to learn more about Earth's past. "What was plausible, what was not plausible," Catling says. Thirteen years ago the Simons Foundation started a big origins project, and Catling jumped onboard a few years later. The private money, he says, might have doubled the amount available for OoL studies, which are sometimes neglected because they overlap disciplines. But, as the Simons project demonstrates, attempts to understand the origins of life are necessarily interdisciplinary, drawing together the fundamental forces and rules that shape the planet and the universe and that don't, themselves,

recognize academic or social boundaries.

起初，卡特林不确定自己对生命起源研究有多少可贡献之处。“我不是生物化学家。我不是有机合成化学家，”他说。但他了解到，化学家们需要像他这样的人，去做他恰好看兴趣的事：更多地了解地球过去。“什么是看似合理的，什么是看似不合理的，”卡特林说。十三年前，西蒙斯基金会启动了一个大型起源项目，卡特林几年后加入。他说，这笔私人资金或许使生命起源研究可用经费翻了一番——这一领域有时因跨越学科而被忽视。但正如西蒙斯项目所示，理解生命起源的尝试必然是跨学科的，汇聚塑造行星与宇宙的基本力与规则，而这些力与规则本身并不承认学术或社会的边界。

Scientists don't differ just in how they study OoL. Their hypotheses about how life first animated itself are also in tension. These "how's" fall into two main camps, whose names seem designed to start a fight: "Metabolism First" and "RNA First."

科学家们的分歧不止于如何研究生命起源。他们关于生命最初如何自我激活的假说也相互张力。这些“如何”落入两大阵营，其名称仿佛专为挑起争端而设：“代谢优先”与“RNA 优先”。

In the latter, RNA was the initial important part of biology to form. RNA is single-stranded genetic material that carries information in the pattern of its four repeating chemical bases: adenine, cytosine, guanine and uracil. (DNA is made of those first three bases but replaces uracil with thymine and has two twisted-together strands instead of one.)

在后者中，RNA 是生物学中最先形成的重要部分。RNA 是单链遗传物质，以四种重复化学碱基——腺嘌呤、胞嘧啶、鸟嘌呤与尿嘧啶——的排列携带信息。（DNA 由

前三种碱基构成，但以胸腺嘧啶取代尿嘧啶，并且是两条缠绕在一起的链，而非一条。）

In scientists' vision of "RNA World," an early era in life's evolution before DNA or proteins existed, RNA stored information and catalyzed chemical reactions. RNA would have assembled into single strands that could make copies of themselves. That copying, by providing a way to reproduce and evolve, could have then kicked off the beginning of life. The more structurally complex DNA would have come later (in a process scientists don't yet understand) to play the major role in storing and passing down genetic information, and proteins would have taken over the chemical catalyzing.

在科学家所构想的“RNA 世界”——生命演化中 DNA 或蛋白质出现之前的早期时代——RNA 既储存信息，又催化化学反应。RNA 会组装成能自我复制的单链。那种复制通过提供繁殖与演化的途径，便可能启动生命的开端。结构更复杂的 DNA 会后来出现（科学家们尚不理解这一过程），承担储存与传递遗传信息的主要角色，而蛋白质则接管化学催化。

Some questions are human classics. Why are we here? Are we alone? What happens when we die? Where do we come from?

有些问题是人类的经典。我们为何在此？我们是否孤独？死后会发生什么？我们从何处来？

The Metabolism First idea, meanwhile, suggests the critical first development was the ability to convert basic chemicals into more complex compounds. According to this scenario, chemical reactions happened spontaneously between elements and compounds on the early Earth. Those

began simply, but some happened to produce energy-storing molecules. Those molecules started more chemical reactions, which became increasingly complex as the cycle continued. Eventually the self-sustaining chemical reactions resulted in structures resembling RNA. Or at least that's the idea.

与此同时，“代谢优先”的想法认为，关键的第一步发展是把基本化学物质转化为更复杂化合物的能力。按此情景，早期地球上的元素与化合物之间自发发生化学反应。起初很简单，但有些碰巧产生了储存能量的分子。那些分子启动了更多化学反应，循环继续，复杂性递增。最终，自我维持的化学反应产生了类似 RNA 的结构。至少，这是想法。

The location of either of those firsts is the source of yet another debate. Some scientists think ground zero was in Earth's surface water: a warm little pond (an idea Charles Darwin first floated) or a hot spring; others think it may have happened in a deep-sea hydrothermal vent, although this hypothesis is falling out of fashion. The heat common to all these ideas would have spurred chemical reactions along.

这两种“优先”发生的地点，又是另一场争论的源头。一些科学家认为零点在地球表层水体：温暖的小池塘（查尔斯·达尔文最先提出的想法）或温泉；另一些人认为可能发生在深海热液喷口，尽管这一假说正日渐失宠。所有这些想法中共同的热，会推动化学反应前行。

Another waning idea is the guess that space rocks may have seeded life on this planet. Samples of asteroids and meteorites suggest that although they have organic molecules, they don't have the kind whose structure matches what life uses here. "Clearly, there's something different that

you need for the environment, for an origin of life,” Catling says. “And it worked on a planetary body like Earth, but it didn’t work elsewhere in the solar system.”

另一个正在消退的想法是：太空岩石可能将生命播种到这颗行星上。对小行星与陨石样本的研究表明，尽管它们含有有机分子，却并不含有结构与地球生命所用相匹配的那种。“显然，起源生命对环境还有别的要求，”卡特林说。“它在像地球这样的行星体上奏效了，但在太阳系其他地方没有。”

He suspects the difference is that asteroids don’t have the same environmental conditions found on Earth—or on the worlds inside the Planet Simulator. One example is wet-dry cycles: in a pond or on the simulator’s chip, for instance, water may evaporate or freeze, leaving behind non-H₂O building blocks that then become more concentrated. Energy from the sun helps reactions between those blocks along, coaxing them to form polymers, or longer chains of molecules.

他怀疑区别在于：小行星不具备地球上——或行星模拟器里那些世界中——相同的环境条件。一个例子是干湿循环：在池塘或模拟器芯片上，水可能蒸发或结冰，留下非 H₂O 的构件，随后变得更浓缩。来自太阳的能量帮助这些构件之间的反应，诱使它们形成聚合物——更长的分子链。

Hydrothermal vents as an origin site are also losing popularity at the moment for a similar reason, in Catling’s view: surrounded by the entire ocean, famously large, vents could not have concentrated chemical constituents enough.

在卡特林看来，热液喷口作为起源地点眼下也正失去人气，理由类似：被整个海洋——那著名的庞然大物——

包围，喷口无法把化学组分浓缩到足够程度。

No matter where a scientist falls among the OoL options—which method, what first and where—there’s still a fundamental problem: When researchers such as Rheinstädter re-create conditions from the early Earth, the ingredients readily assemble themselves into the simpler building blocks of life, among them amino acids or even cell compartments, which are made of fats called lipids. But getting those building blocks to stick together and interact in a lively way is a different story. “What’s still very much a dark field is how those things become more complex,” says Joana Xavier, an independent bioengineer and OoL researcher. Rheinstädter agrees. “I think what we have right now is that the building blocks—amino acids, nucleotides—they form very easily,” he says. “Even if you step out in a puddle, you will find those.”

无论一位科学家落在生命起源选项的哪一边——何种方法、何者优先、何处发生——仍有一个根本问题：当莱因施泰特这类研究者重现早期地球条件时，原料很容易自行组装成生命较简单的构件，其中包括氨基酸乃至由称为脂质的脂肪构成的细胞隔室。但让这些构件粘在一起并以活泼的方式相互作用，则是另一回事。“如何让那些东西变得更复杂，仍然是一片非常黑暗的领域，”独立生物工程师、生命起源研究者若安娜·哈维尔说。莱因施泰特表示同意。“我认为我们现在拥有的是：构件——氨基酸、核苷酸——形成得非常容易，”他说。“即使你踏进一个水坑，也能找到那些东西。”

Rheinstädter is using the Planet Simulator to figure out how to get those puddle parts to complexify themselves and, in particular, assemble into RNA. In that quest, he has a fairly radical idea: Maybe the chemistry soups re-

searchers are creating are getting sufficiently complex—but scientists are missing it because they're seeking the wrong thing. "People are always looking for lab-grade RNA," he says—the kind you can order from a catalog. Perfect, in other words.

莱因施泰特正用行星模拟器弄清如何让那些水坑零件自我复杂化，尤其是组装成 RNA。在这一求索中，他有一个相当激进的想法：或许研究者们制造的化学汤已经足够复杂——但科学家们错过了它，因为他们在寻找错误的东西。“人们总是在找实验室级的 RNA，”他说——那种可以从目录订购的。换句话说，完美的。

But that kind of RNA wouldn't have done well on the four-billion-year-old Earth. It's too fragile. "If you take it out of the fridge, it degrades immediately," Rheinstädter says. Maybe, he continues, that means the RNA in RNA World wouldn't have been the same as the RNA in our world. He came to this idea somewhat by accident, while poking around the Planet Simulator. When the simulator was set to mimic the ancient Earth, Rheinstädter and his team found that molecules do link up to form RNA strands of about 100 bases long. "However, they're not perfect," he says. "They have structural defects." But what they lack in prettiness they make up for in strength. "You cannot destroy them," Rheinstädter says.

但那种 RNA 在四十亿年前的地球上不会表现得好。它太脆弱了。“你把它从冰箱拿出来，立刻就降解，”莱因施泰特说。或许，他继续说，这意味着 RNA 世界中的 RNA 不会与我们世界中的 RNA 相同。他几乎是偶然想到这一点的，当时他在行星模拟器里摸索。当模拟器被设定为模仿远古地球时，莱因施泰特及其团队发现分子确实连接成约 100 个碱基长的 RNA 链。“然而，它们并不完美，”他说。“它们有结构缺陷。”但它们在美观上的

不足，由强度弥补。“你无法摧毁它们，”莱因施泰特说。

He started to wonder: "What if nature didn't start off perfect?" What if, instead, it started out with a precursor version of genetic material? The RNA of today may have come later, with evolution. "Maybe it's just silly to expect that from the beginning," he says. Perhaps the mixed-up version of RNA that spews from the Planet Simulator was the kind that first bubbled up from some terrestrial pond. After all, what—besides a creation story—starts out with a fully formed version of life?

他开始思索：“如果自然一开始就不完美呢？”如果，相反，它始于遗传物质的某种前体版本呢？今天的 RNA 或许是后来伴随演化而来的。“或许从一开始就期待那个，本身就很傻，”他说。也许从行星模拟器喷涌而出的那种混杂版 RNA，才是最初从某个陆地池塘冒出来的那种。毕竟——除了创世故事——有什么是以完全成形的生命版本开场的呢？

Other scientists, Rheinstädter posits, may have actually been creating this flawed RNA all along in similar experiments, unaware—because they weren't looking for it. Now the team is working on whether imperfect RNA can actually do what perfect RNA does: catalyze and transmit information.

莱因施泰特猜想，其他科学家或许一直在类似实验中制造这种有缺陷的 RNA，却浑然不觉——因为他们没有在找它。如今团队正研究：不完美的 RNA 是否真能做完美 RNA 所做的事：催化并传递信息。

Catling, meanwhile, is pursuing a shakeup idea of his own involving the "where" debate. He's been looking recently at what scientists call the "phosphate problem," first identified in the 1950s. The issue: Phosphate is key to many

things life does and needs—the compound is part of the genome, the cell compartment and metabolism, for instance. ”Yet in the environment, if you go to a river or the ocean, the level of dissolved phosphate is really tiny,” Catling says.

与此同时，卡特林正就“何处”之争推进他自己的颠覆性想法。他最近在关注科学家所称的“磷酸盐问题”，这一问题最早于 1950 年代被指出。问题在于：磷酸盐对生命所做与所需的许多事情至关重要——例如，这种化合物是基因组、细胞隔室与代谢的一部分。“然而在环境中，如果你去到一条河或海洋，溶解磷酸盐的水平真的非常低，”卡特林说。

If there's not much in the water, it's hard to understand how proto-life could have gathered enough. But when Catling was working on the Simons project, he and colleagues found a kind of water body with plenty of phosphate: soda lakes, which are a bit like big ponds someone dissolved a lot of baking soda into. These water bodies with an abundance of carbonate and extremely high alkalinity are found around the world; the best-known examples are in Africa and North America. The calcium in them binds with carbonate (the baking soda bit), allowing natural phosphates to accumulate instead of sticking to calcium as they would in less carbonate-rich water.

若水中含量不多，就很难理解原生命如何聚集到足够的量。但当卡特林参与西蒙斯项目时，他与同事发现一类富含磷酸盐的水体：苏打湖，有点像有人把大量小苏打溶进去的大池塘。这些碳酸盐丰富、碱度极高的水体遍布世界；最著名的例子在非洲与北美。其中的钙与碳酸盐（小苏打那部分）结合，使天然磷酸盐得以积累，而不是像在碳酸盐较少的水中那样粘在钙上。

The most phosphate-rich lake in the world is Last Chance Lake in Canada. In summertime, it—and most soda lakes—evaporates, leaving a crusty white landscape; as the seasons progress, they get wet and become like lakes again. The wet-dry cycle concentrates the existing phosphate. It could, Catling thinks, be precisely the type of place where chemistry and physics became biology. Maybe it would more appropriately be called First Chance Lake.

世界上磷酸盐最丰富的湖是加拿大的最后机会湖。夏天，它——以及多数苏打湖——蒸发，留下一片硬壳般的白色景观；随着季节推进，它们又变湿，重新像湖。干湿循环浓缩了既有的磷酸盐。卡特林认为，它恰恰可能是化学与物理变成生物学的那类地方。或许更该叫作“第一次机会湖”。

Of course, neither Rheinstädter nor Catling knows if their ideas represent the true “first” or “where.” And scientists disagree about whether we will ever know for sure. Even if researchers can reproduce the conditions of early Earth and coax life from it, that success would represent just one way the story might have gone. Knowing for sure that it is the way it happened is a different story. The tape can never be rewound.

当然，莱因施泰特与卡特林都不知道自己的想法是否代表真正的“优先”或“何处”。科学家们对于我们是否终将确知也意见不一。即便研究者能重现早期地球条件并诱使生命从中出现，那一成功也只代表故事可能走过的一条路。确知那就是实际发生的路，则是另一回事。磁带永远无法倒带。

Plenty of OoL scientists, though, think that doubt is naive. After all, if cellular biology flows from physics and chemistry through a set of logical laws and processes, maybe

there is only one path between the three. "A diamond forms in a very specific set of conditions. It's only in that set of conditions," Xavier says. "And the diamond is rare because those conditions happen only on those rare occasions. I think the formation of cells is exactly the same type of question."

不过，许多生命起源科学家认为这种怀疑是幼稚的。毕竟，若细胞生物学经由一套逻辑定律与过程从物理与化学流出，那么三者之间或许只有一条路径。“钻石在非常特定的一组条件下形成。只有在那组条件下，”哈维尔说。“钻石之所以稀有，是因为那些条件只在罕见场合出现。我认为细胞的形成正是同一类问题。”

Catling, though, disagrees. "I think it would be a plausible answer rather than definitive." He does, though, think that looking beyond Earth could yield more certainty, in that it could help scientists understand whether the origin of cellular life is actually deterministic—that is, whether, given certain conditions, it always emerges in a certain way. If, for instance, we see it on other planets with similar environmental histories and if it has the same biochemical makeup, then researchers could infer that there's a set process by which life like ours arises.

卡特林则不同意。“我认为那会是一个貌似合理的答案，而非确定的答案。”不过，他认为把目光投向地球之外可能带来更多确定性，因为这能帮助科学家理解细胞生命的起源是否确实是决定论的——也就是说，在给定某些条件下，它是否总以某种方式出现。例如，若我们在环境史相似的其他行星上看到它，且它具有相同的生物化学构成，那么研究者便可推断：像我们这样的生命有一套既定的产生过程。

Still, those are huge ifs: If there's life out there, if we know

how and where to look for it, and if we find it. That's a lot of physical and philosophical difficulty to bring together. But it's also not impossible. "It's always dangerous in the history of science to say that we will never know something, because later on, people have found it," says David Dunér, a historian and philosopher of science at Lund University in Sweden.

不过，那是巨大的“如果”：如果外面有生命，如果我们知道如何、何处去找，如果我们找到了它。那要把大量物理与哲学上的困难汇到一处。但这也并非不可能。“在科学史上说我们永远不会知道某事，总是危险的，因为后来人们找到了它，”瑞典隆德大学科学史与科学哲学家戴维·杜内尔说。

Before the field of OoL can get close to that point, though, it needs a cultural change. At the moment, it's a fairly contentious discipline, with researchers personally invested in their versions of how and what first and where. This competition is amplified by how little direct evidence is available, Xavier contended in a *Nature* comment. To help researchers collaborate and focus on the big picture, she co-founded the Origin of Life Early-career Network, bringing together younger scientists. But schisms and heel digging are probably inevitable in such uncertain fields, Catling says.

然而，在生命起源领域接近那一点之前，它需要一场文化变革。眼下，这是一个相当充满争论的学科，研究者们个人投入于各自版本的如何、何者优先与何处。直接证据稀少又放大了这种竞争，哈维尔在《自然》的一篇评论中指出。为帮助研究者协作并聚焦大局，她共同创立了生命起源早期职业网络，把年轻科学家聚到一起。但卡特林说，在如此不确定的领域，分裂与固执大概不可避免。

Maybe more evidence for the origin of life will come from a soda lake, or a planet simulator, or an actual planet far, far away. But at least some scientists are okay with the uncertainty about uncertainty. "I sometimes think about how lucky I am, in some ways, to exist in this moment of time when this still is an unanswered question, and how exciting it is to think about all the possibilities out there," Smith says. "I think that it will be answered, but I think it will be some time because a lot of the right questions aren't even articulated."

也许更多关于生命起源的证据将来自一座苏打湖，或一台行星模拟器，或一颗遥远的真实行星。但至少有些科学家对“关于不确定性的不确定性”安之若素。“我有时会想，在某种意义上，我多么幸运能存在于这个这个问题仍未解答的时刻，去思索外面所有的可能性又是多么令人兴奋，”史密斯说。“我认为它会被解答，但我认为还需要一些时间，因为许多正确的问题甚至还没有被表述出来。”

Will we ever achieve immortality?

人类能否永生？

Will we ever achieve immortality?

By Venki Ramakrishnan · September 15, 2026 · edited by Clara Moskowitz

<https://www.scientificamerican.com/article/can-humans-live>

Scientists are pursuing the limits of longevity

科学家们正追寻长寿的极限

Humans may be the only creatures that are aware of their mortality. There is no evidence that other animals know their time on Earth is finite. The realization that we and everyone we know will someday die must be one of the more unsettling discoveries of childhood.

人类或许是唯一意识到自身有死的生物。没有证据表明其他动物知道它们在地球上的时间是有限的。认识到我们以及我们认识的每一个人终将死去，想必是童年更令人不安的发现之一。

Ever since humans recognized we have a limited lifespan, that knowledge has shaped our cultures, religions and philosophies. Although we make plans, pursue careers and behave as though we will live indefinitely, the fear of death and the desire to transcend it remain.

自从人类认识到寿命有限，这一认知便塑造了我们的文化、宗教与哲学。尽管我们制订计划、追求事业，行为举止仿佛将永远活下去，但对死亡的恐惧与超越死亡的渴望依然存在。

As a species, we have made some major strides against mortality. In modern times, death from violence, disease and famine has become less common than it once was. At the same time, medical advances and a dramatic decline in fertility rates have produced aging societies. The proportion of people older than 65 is rising rapidly, while the working-age population that supports them shrinks. Understanding aging is therefore an urgent practical challenge. If we can delay its effects, older people may remain healthy, independent and productive for longer.

作为物种，我们已在对抗死亡方面迈出重大步伐。在现时代，暴力、疾病与饥荒造成的死亡已不如从前常见。与此同时，医学进步与生育率的急剧下降造就了老龄化社会。65 岁以上人口比例迅速上升，而支撑他们的劳动年龄人口却在萎缩。因此，理解衰老是一项紧迫的现实挑战。若我们能延缓其效应，老年人或许能更长久地保持健康、独立与生产力。

But alongside this public health imperative is an enduring dream. A growing number of tech billionaires are funding efforts not merely to slow aging but to defeat it altogether. Their goal is radical life extension and, ultimately, something approaching immortality. But is that even possible? These efforts have scientists chasing one of the biggest mysteries of life itself: Is there a biological reason that humans must die? And if not, could we one day live forever?

但与这一公共卫生要务并存的，是一个经久不衰的梦想。

越来越多的科技亿万富翁出资，不只是为了减缓衰老，而是要彻底战胜它。他们的目标是激进的寿命延长，最终接近某种永生。但这究竟可能吗？这些努力让科学家们追逐生命本身最大的谜题之一：人类是否必须死去，有没有生物学上的理由？若没有，我们有朝一日能永远活下去吗？

The diversity of lifespans in nature suggests that aging isn't fixed. Despite being constructed from the same chemical building blocks, species can have dramatically different lengths of life. Mayflies may live only days, mice about two years, and bowhead whales and Greenland sharks for centuries. Galapagos tortoises live so long that one of them hobbling around today could have encountered Charles Darwin in 1835.

自然界寿命的多样性表明，衰老并非固定不变。尽管由相同的化学构件组成，不同物种的寿命可以截然不同。蜉蝣也许只活几天，老鼠大约两年，弓头鲸与格陵兰鲨则可达数百年。加拉帕戈斯象龟活得如此之久，今天蹒跚而行的其中一只，或许在 1835 年遇见过查尔斯·达尔文。

These differences are not random. Lifespan is closely related to metabolism. Smaller animals generally have higher metabolic rates and shorter lives; larger animals tend to metabolize energy more slowly and live longer.

这些差异并非随机。寿命与代谢密切相关。体型较小的动物通常代谢率较高、寿命较短；体型较大的动物往往代谢能量更慢、活得更久。

Natural selection doesn't care how long an organism lives. It favors traits that increase reproductive success. Every species makes trade-offs in how it allocates resources among growth, reproduction and maintenance. A mouse

doesn't improve its ability to produce more offspring by investing in longevity if starvation or predators are likely to kill it first. For a mouse, investing in rapid growth and reproduction is more advantageous. But larger animals, such as whales, mature slowly and reproduce later. Their bigger size also reduces the risk of predation, making long-term maintenance more worthwhile.

自然选择并不在意一个有机体活多久。它青睐能提高生殖成功的性状。每个物种都在生长、繁殖与维持之间如何分配资源上做出权衡。若饥饿或捕食者很可能先杀死一只老鼠，它通过投资长寿来提高后代产量并无益处。对老鼠而言，投资于快速生长与繁殖更有利。但像鲸鱼这样的大型动物成熟缓慢、繁殖较晚。更大的体型也降低了被捕食的风险，使长期维持更值得。

The same logic explains some notable exceptions. Birds and bats often live much longer than similarly sized terrestrial mammals because flight reduces the risk of predation and also allows them to forage over a wide area, cutting the chances of starvation. When surviving longer increases reproductive success, evolution favors slower aging.

同样的逻辑解释了一些显著例外。鸟类与蝙蝠往往比体型相近的陆生哺乳动物活得久得多，因为飞行降低了被捕食风险，也使它们能在广阔区域觅食，减少饿死的几率。当活得更久能提高生殖成功时，演化便青睐更慢的衰老。

Evolution can also reward genes that are beneficial early in life even when they become harmful later. For example, the same genes that promote rapid growth or prevent cancer during youth can contribute to aging later on. This evolutionary theory of aging is known as antagonis-

tic pleiotropy. "Pleiotropy" simply means a gene can have more than one effect, and "antagonistic" means some of those effects can be harmful. The theory suggests that aging is not an adaptation in itself but an unintended consequence of evolutionary compromises. The upshot is that each species has a maximum natural lifespan.

演化也可能奖励那些在生命早期有益、即便日后有害的基因。例如，促进快速生长或在青年期预防癌症的同一基因，日后可能促成衰老。这一衰老的演化理论被称为拮抗性多效性。“多效性”只是说一个基因可有不止一种效应，“拮抗”则意味着其中一些效应可能有害。该理论表明，衰老本身并非一种适应，而是演化妥协的意外后果。结论是：每个物种都有一个最大自然寿命。

Human life expectancy has roughly doubled over the past 150 years.

过去 150 年间，人类预期寿命大约翻了一番。

What about us? Human life expectancy has roughly doubled over the past 150 years because of improvements in public health, nutrition and medicine. Maximum lifespan is a different matter. Even centuries ago some individuals lived remarkably long lives. Michelangelo lived to be almost 90 years old. The longest documented human life is that of Jeanne Calment, who died at 122 in 1997. No one since then has lived past 120. Under current biological constraints, humans appear unlikely to live much longer than that. Moving much beyond it would require altering the biology of aging itself.

那我们呢？由于公共卫生、营养与医学的改善，过去 150 年间人类预期寿命大约翻了一番。最大寿命则是另一回事。即便数百年前，也有人活得格外长久。米开朗琪罗

活到了将近 90 岁。有文献记载的最长人类寿命是让娜·卡尔芒，她于 1997 年以 122 岁去世。此后无人活过 120 岁。在当前生物学约束下，人类似乎不太可能活得比那长太多。要大幅超越，就需要改变衰老本身的生物学。

As people age, our muscle mass declines, reducing strength, mobility and balance. We suffer from fatigue and exhaustion and have difficulty carrying out everyday activities. Chronic diseases become more common, and resistance to infection weakens. Cognitive function declines as well. These symptoms are merely external manifestations of a gradual loss of cellular and tissue function, which themselves are a result of changes and damage to the molecules of life and their regulation by our genes.

随着人们变老，肌肉量下降，力量、活动能力与平衡减弱。我们遭受疲劳与衰竭，难以完成日常活动。慢性病更常见，对感染的抵抗力减弱。认知功能也下降。这些症状不过是细胞与组织功能逐渐丧失的外在表现，而其本身则是生命分子及其受基因调控的变化与损伤的结果。

Aging was once viewed as simple wear and tear, like a machine slowly breaking down, but modern biology has revealed a more complicated picture. Cells possess sophisticated systems for repairing damage, maintaining our repertoire of proteins, recycling components and regulating metabolism. Aging occurs not merely because damage accumulates but because these protective systems themselves become less effective over time.

衰老一度被视为简单的磨损，像一台机器慢慢坏掉，但现代生物学揭示了一幅更复杂的图景。细胞拥有修复损伤、维持蛋白质库、回收组分与调节代谢的精密系统。衰老的发生，不只是因为损伤积累，也因为这些保护系统本身随时间变得不那么有效。

The hallmarks of biological aging appear in multiple body systems. They include damage or changes to our DNA, deterioration of the quality and distribution of proteins, dysfunction of mitochondria, decline in our response to external nutrients, chronic inflammation, and reductions in stem cells. These systems are deeply interconnected—a disturbance in one often affects many others.

生物衰老的标志出现在多个身体系统中。它们包括 DNA 的损伤或改变、蛋白质质量与分布的恶化、线粒体功能障碍、对外界营养响应的衰退、慢性炎症，以及干细胞减少。这些系统深度互联——一处扰动往往影响许多其他系统。

The encouraging news is that there is no physical or chemical law preventing us from modifying these processes. The challenge is that aging is extraordinarily complex.

令人鼓舞的消息是：没有物理或化学定律阻止我们修改这些过程。挑战在于衰老异常复杂。

Among the most powerful interventions discovered so far is caloric restriction. Animals given just enough calories to maintain adequate nutrition often remain healthier and live significantly longer. Caloric restriction reduces protein synthesis and stimulates autophagy, the cellular recycling process that removes damaged molecules and organelles. In species ranging from worms and flies to rodents and primates, caloric restriction produces animals that appear biologically younger than their age. The drawbacks, however, are considerable. Constant hunger, sensitivity to cold, impaired wound healing and reduced libido make strict caloric restriction difficult to sustain.

迄今发现的最有力干预之一是热量限制。给予刚好足以维持充足营养的热量的动物，往往更健康，活得显著更

久。热量限制减少蛋白质合成，并刺激自噬——清除受损分子与细胞器的细胞回收过程。从蠕虫、果蝇到啮齿类与灵长类，热量限制产生的动物看起来比实际年龄更年轻。然而弊端相当可观。持续饥饿、怕冷、伤口愈合受损、性欲下降，使严格热量限制难以维持。

Scientists have searched for drugs that mimic some of the effects of caloric restriction without the side effects. One that appears promising is rapamycin, originally discovered in soil samples from Easter Island, also known as Rapa Nui by its native people. This compound affects a cellular pathway called TOR (or mTOR) that is involved in growth and metabolism. In mice, it extends lifespan and improves several measures of health. Yet rapamycin is also an immunosuppressant, raising concerns about infection and long-term safety. Current research focuses on whether lowering the dose or synthesizing chemical analogs of rapamycin can produce its beneficial effects without the problems.

科学家们一直在寻找能模仿热量限制某些效应而无其副作用的药物。一种看似有前景的是雷帕霉素，最初发现于复活节岛（当地人亦称拉帕努伊）的土壤样本。这种化合物影响一条称为 TOR（或 mTOR）的细胞通路，该通路参与生长与代谢。在小鼠中，它延长寿命并改善若干健康指标。但雷帕霉素也是免疫抑制剂，引发对感染与长期安全性的担忧。当前研究聚焦于：降低剂量或合成雷帕霉素的化学类似物，能否在没有那些问题的情况下产生其有益效应。

Metformin, a widely used diabetes drug, has likewise attracted interest. It acts on a protein in mitochondria involved in respiration, which indirectly influences caloric-restriction pathways. Evidence for benefits in healthy in-

dividuals, though, is mixed. The recent GLP-1 drugs developed for diabetes and obesity also appear to have beneficial effects on multiple organs, including the brain. Yet they, too, have side effects, including muscle loss. For use as antiaging therapies, both metformin and GLP-1 drugs will require careful clinical trials.

广泛使用的糖尿病药物二甲双胍同样引起关注。它作用于线粒体中参与呼吸的一种蛋白，间接影响热量限制通路。不过，对健康个体益处的证据参差不齐。近来为糖尿病与肥胖开发的 GLP-1 药物，似乎也对包括大脑在内的多个器官有益。但它们同样有副作用，包括肌肉流失。若要作为抗衰老疗法使用，二甲双胍与 GLP-1 药物都需要审慎的临床试验。

Another major target of antiaging science is cellular senescence. Cells experiencing damage or stress sometimes enter a senescent state in which they stop dividing and begin secreting inflammatory molecules. Early in life this response is beneficial, aiding development, wound healing and cancer prevention. With age, however, senescent cells accumulate faster than the body can remove them.

抗衰老科学的另一大靶点是细胞衰老。经历损伤或应激的细胞有时进入衰老状态：停止分裂，并开始分泌炎症分子。生命早期，这一响应是有益的，有助于发育、伤口愈合与癌症预防。然而随着年龄增长，衰老细胞的积累速度快于身体清除它们的速度。

The resulting inflammation contributes to tissue dysfunction throughout the body. In mice, eliminating senescent cells can improve many features of aging. This finding has sparked intense interest in so-called senolytic drugs designed to remove such cells selectively. Several candidates are now being tested against age-related diseases such as

osteoarthritis and fibrosis. Apart from making sure the drugs are safe, the challenge is achieving the right balance: senescent cells serve a biological purpose, and eliminating them indiscriminately could lead to new problems.

由此产生的炎症促成全身组织功能障碍。在小鼠中，清除衰老细胞可改善衰老的许多特征。这一发现激起了对所谓“衰老细胞清除剂”（senolytic）药物的浓厚兴趣，这类药物旨在选择性清除此类细胞。若干候选药物正针对骨关节炎与纤维化等年龄相关疾病进行测试。除确保药物安全外，挑战在于达到正确平衡：衰老细胞有生物学用途，不加区分地清除它们可能导致新问题。

Another, somewhat bizarre experiment has raised the prospect of a different antiaging strategy. In a process called parabiosis, researchers physically joined pairs of rats so they shared circulatory systems. Old rats paired with younger ones outlived those paired with animals their own age. Such findings went on to inspire a slew of companies offering transfusions of plasma from young donors. There is no convincing evidence that such procedures slow aging in humans. Nevertheless, the work has set off a major search for factors in blood that change with age and may eventually prove therapeutically useful.

另一项略显怪异的实验，提出了不同抗衰老策略的前景。在一种称为并体共生的过程中，研究者把成对的大鼠身体连接起来，使它们共享循环系统。与年轻大鼠配对的年老大鼠，比与同龄动物配对的活得更久。此类发现继而催生了一批提供年轻供体血浆输注的公司。没有令人信服的证据表明此类程序能减缓人类衰老。尽管如此，这项工作引发了对血液中随年龄变化、最终可能具有治疗用途的因子的重大搜寻。

Most of these approaches aim to slow down aging. A more

radical goal is to reverse it.

这些方法大多旨在减缓衰老。更激进的目标是逆转衰老。

This possibility emerged from a series of discoveries, spanning decades, involving the cells of the early embryo, which possess remarkable flexibility. These so-called pluripotent stem cells can generate any tissue in the body. In later stages, the stem cells become progressively more specialized, and each class generates specific types of tissue such as nervous system cells or blood.

这一可能性源自跨越数十年的一系列发现，涉及早期胚胎细胞，它们拥有非凡的灵活性。这些所谓的多能干细胞能生成体内任何组织。在后续阶段，干细胞逐渐特化，每一类生成特定类型的组织，如神经系统细胞或血液。

In 1958 British biologist John Gurdon showed that the nucleus from a cell in a tadpole could be transplanted into an egg cell to create an entirely new frog. Later, in 1997, Dolly the sheep served as a demonstration that cloning was possible in mammals. The exact process for reverting adult cells back to an early embryonic stage, however, wasn't clear.

1958 年，英国生物学家约翰·格登表明，可将蝌蚪细胞的细胞核移植到卵细胞中，创造出一只全新的青蛙。后来，在 1997 年，多莉羊证明了哺乳动物克隆是可能的。然而，将成体细胞恢复到早期胚胎阶段的确切过程当时并不清楚。

A major breakthrough came in 2006, when Japanese biologist Shinya Yamanaka discovered that activating just four genes could convert adult cells into pluripotent stem cells. These Yamanaka factors effectively induce cells to return to an earlier embryonic stage by genetically reprogramming them. The discovery raised the possibility that

if we can reset cellular age, perhaps we can reverse aging itself.

重大突破出现在 2006 年：日本生物学家山中伸弥发现，仅激活四个基因就能把成体细胞转化为多能干细胞。这些山中因子通过基因重编程，有效诱导细胞回到较早的胚胎阶段。这一发现提出了一种可能：若我们能重置细胞年龄，或许就能逆转衰老本身。

Applying the full reprogramming process to an adult organism would be dangerous because the resulting pluripotent cells could form tumors known as teratomas. Researchers therefore began exploring partial reprogramming, briefly activating the Yamanaka factors in ways that appear to rejuvenate cells while maintaining their specialized identities.

将完整重编程过程应用于成体有机体将是危险的，因为所得的多能细胞可能形成称为畸胎瘤的肿瘤。因此研究者开始探索部分重编程：短暂激活山中因子，以看似能在维持特化身份的同时使细胞年轻化的方式进行。

Results in mice have been striking. Treated animals often show improvements in tissue function and markers of biological age. In one experiment, scientists reportedly regenerated a damaged optic nerve by injecting Yamanaka factors via a viral carrier into the eye. Researchers are now conducting clinical trials to see whether the procedure can reverse degeneration in the retina or optic nerve.

小鼠结果令人瞩目。接受处理的动物往往在组织功能与生物年龄标志上有改善。在一项实验中，科学家据报通过病毒载体向眼内注射山中因子，再生了受损的视神经。研究者们现正开展临床试验，以查看该程序能否逆转视网膜或视神经的变性。

Most antiaging approaches aim to slow down the process. A more radical goal is to reverse it.

多数抗衰老方法旨在减缓这一过程。更激进的目标是逆转它。

Yet formidable obstacles remain. Current methods typically rely on gene-delivery systems that are not ideal for widespread human use. Reprogramming is often uneven, affecting clusters of cells and leaving others in the same organ unchanged. Different organs age at different rates and may require specific targeting. Researchers are pursuing safer and more controllable methods, including small molecules that might produce reprogramming without the introduction of new genes.

然而仍有巨大障碍。当前方法通常依赖并不理想地适用于广泛人类使用的基因递送系统。重编程往往不均匀，影响成簇细胞，而同一器官中的其他细胞不变。不同器官以不同速率衰老，可能需要特定靶向。研究者正追求更安全、更可控的方法，包括可能在不引入新基因的情况下产生重编程的小分子。

Despite these challenges, cellular reprogramming represents the most compelling route yet toward genuine rejuvenation, rather than merely slowing decline.

尽管有这些挑战，细胞重编程代表了迄今通向真正年轻化——而非仅仅减缓衰退——的最有说服力的路径。

Even if such interventions succeed, translating them into medicine will be difficult. Antiaging therapies face a higher bar than treatments for life-threatening diseases. Patients with advanced cancer may accept substantial risks in experimental medicines because the alternative is death. Healthy

individuals are unlikely to tolerate serious side effects from drugs they may have to take for decades. And clinical trials present another obstacle. Regulatory agencies generally approve therapies for specific diseases, not for aging itself. As a result, scientists often test antiaging interventions indirectly through research on age-related disorders.

即便此类干预成功，将其转化为医学也将困难。抗衰老疗法面临比危及生命疾病的治疗更高的门槛。晚期癌症患者可能接受实验药物的重大风险，因为替代选项是死亡。健康个体不太可能容忍他们或许要服用数十年的药物的严重副作用。临床试验又是另一障碍。监管机构通常批准针对特定疾病的疗法，而非针对衰老本身。因此，科学家们往往通过年龄相关疾病的研究间接测试抗衰老干预。

Much antiaging research is motivated by the idea of extending "healthspan"—the number of healthy years we enjoy—without necessarily extending lifespan. According to this philosophy, people should aim to remain healthy and independent for longer and then experience a relatively brief period of decline before death. It is an attractive idea, but there is little evidence that it's possible. In many countries, people are living longer, yet the fraction of life spent with multiple chronic illnesses has increased. On the other hand, there are striking differences between individual countries in the proportion of life spent in good health. People in Sweden, for example, have a considerably shorter period of late-life disability than those in the U.K. Such differences suggest that public health policies, lifestyle factors and medical care can influence not only how long we live but how well we live.

许多抗衰老研究的动机是延长“健康寿命”——我们享有健康的年数——而不一定延长寿命。按此哲学，人们

应争取更长久地保持健康与独立，然后在死亡前经历相对短暂的衰退期。这是有吸引力的想法，但几乎没有证据表明它可行。在许多国家，人们活得更久了，但生命中患多种慢性病的比例却增加了。另一方面，各国在健康生活所占比例上有显著差异。例如，瑞典人晚年失能期明显短于英国人。此类差异表明，公共卫生政策、生活方式因素与医疗护理不仅能影响我们活多久，也能影响我们活得多好。

Ultimately, though, will any of these findings allow us to significantly extend the human lifespan? There is no scientific law against the idea, but nor is there any reason interstellar travel must be impossible. It's just that the practical complexities are immense. Aging involves countless interacting processes operating across cells, tissues and organs. Extending lifespan dramatically would require mastering many of them simultaneously. It is unlikely that a single intervention could simply switch aging off without affecting us in other ways.

最终，这些发现中是否有任何一项能让我们显著延长人类寿命？没有科学定律反对这一想法，但也没有理由说星际旅行必然不可能。只是现实复杂性巨大。衰老涉及跨越细胞、组织与器官的无数相互作用过程。要大幅延长寿命，需要同时掌握其中许多过程。不太可能有单一干预能简单地关掉衰老而不以其他方式影响我们。

In the meantime, as we await major medical advances, basic strategies allow us all to play the antiaging game on a personal scale. Scientists know a moderate diet with a healthy intake of fruits and vegetables affects caloric-restriction pathways. Much of our body's repair and recycling work is carried out during our sleep cycle, and exercise helps to regenerate tissue, including muscle and mi-

tochondria. For now, the best-supported tools for healthy aging remain the trio of diet, exercise and sleep. In addition, there is good evidence that avoiding social isolation and having a sense of purpose are associated with better health and longevity.

与此同时，在我们等待重大医学进步之际，基本策略让我们都能在个人尺度上玩抗衰老游戏。科学家们知道，含有充足蔬果摄入的适度饮食会影响热量限制通路。身体的大量修复与回收工作在睡眠周期中进行，而运动有助于再生组织，包括肌肉与线粒体。就目前而言，健康衰老得到最多支持的工具仍是饮食、运动与睡眠三件套。此外，有充分证据表明，避免社会隔离并拥有目标感，与更好的健康与长寿相关。

As long as accidents, natural disasters, disease and wars can kill us, true immortality is out of the question. And living for hundreds of years is unlikely anytime soon. Instead we can think of life as an eternal banquet that individuals enter and eventually leave. We should consider ourselves lucky to be part of it at all and enjoy the feast while we can.

只要事故、自然灾害、疾病与战争仍能杀死我们，真正的永生就无从谈起。而活上数百年，短期内也不太可能。相反，我们可以把生命想成一场永恒的宴会，个体进入，最终离开。我们应庆幸自己至少能参与其中，并在还能享用时好好享受这场盛宴。

What if we never find dark matter?

如果我们永远找不到暗物质？

What if we never find dark matter?

By Joseph Howlett · September 15, 2026 · edited by Seth Fletcher

<https://www.scientificamerican.com/article/what-if-we-never-find-dark-matter/>

Could the hunt for the universe's missing matter be the biggest wild goose chase in scientific history?

对宇宙缺失物质的搜寻，会不会是科学史上最大的徒劳追逐？

I used to have a more interesting morning commute. At daybreak I'd squeeze into a tiny Fiat with all the physicists it could fit. We'd drive through winding Italian roads and onto a highway that cuts across the peninsula, with a tunnel right through its grandest mountain, Gran Sasso. At the tunnel's deepest point, with nearly a mile of bedrock resting above our heads, we'd turn off through a guarded gate and enter one of the quietest places on Earth.

我曾有过更有趣的早间通勤。破晓时分，我会挤进一辆小菲亚特，塞满它能装下的所有物理学家。我们穿过蜿蜒的意大利公路，驶上横贯半岛的高速，隧道径直穿过它最宏伟的山峰——大萨索山。在隧道最深处，头顶压

着近一英里的基岩，我们拐进一道有人把守的大门，进入地球上最安静的地方之一。

We were there, in 2019, to build a particle detector more sensitive than any in history. We spent long workdays, overnight shifts and most weekends in this cavern, locked in a race with international rivals to find the missing five sixths of the universe's matter.

2019 年，我们在那里建造有史以来最灵敏的粒子探测器。我们在这洞穴里度过漫长的工作日、通宵班次与多数周末，与国际对手赛跑，寻找宇宙物质中缺失的六分之五。

In spare moments while turning a wrench or staring at a pressure gauge, I'd sometimes ask my teammates what our chances were of winning that race. "Zero," they'd usually reply. Surprised, I began questioning colleagues beyond my several hundred collaborators, polling them at conferences and happy hours, and found that it wasn't just us. Few experts—among the younger generation basically no one—thought any scientific effort was on the brink of discovering dark matter.

在拧扳手或盯着压力计的间隙，我有时会问队友我们赢得那场比赛的几率。“零，”他们通常回答。惊讶之下，我开始在数百名合作者之外追问同事，在会议与欢乐时光里做民调，发现不只是我们。很少有专家——在更年轻的一代中基本上没人——认为任何科学努力正濒临发现暗物质。

In the seven years since, these murmurs have grown only louder. For four decades now we've failed to identify the particles that make up dark matter—the invisible chunk of the universe that we know is there, that suffuses every galaxy we see. And more physicists than ever are

conceding that this situation might never change. "Nature doesn't always promise us the results we'd like," says physicist and mathematician Edward Witten, whose 1985 paper proposing that weakly interacting massive particles, or WIMPs, could be observed on Earth launched the field that I focused my Ph.D. work on. "It's unfortunate, but it's certainly possible that nature chose a dark matter particle that we won't be able to detect," Witten observes.

此后七年，这些窃窃私语只变得更响。四十年来，我们未能鉴定构成暗物质的粒子——那块我们知道存在、弥漫于我们所见每一个星系的不可见宇宙块。越来越多的物理学家承认，这种局面或许永远不会改变。“自然并不总是许诺我们想要的结果，”物理学家兼数学家爱德华·威滕说；他 1985 年提出弱相互作用大质量粒子（WIMP）可在地球上被观测的论文，开启了我博士工作所聚焦的领域。“这很不幸，但自然完全可能选择了一种我们将无法探测的暗物质粒子，”威滕观察道。

Many believers remain. Dark matter hunters are loping desperately in a host of directions both old and new, hoping that nature can't keep this secret forever. At the beginning of September, in a sign of just how desperate the search has become, members of the community lost their minds at the announcement of a single unexplained collision in one of dozens of detectors that didn't even match the WIMP profile. I'd wager \$1,000 that this potential dark matter detection, like every other tiny shred of hope nature has dealt us in this 30-year search, will evaporate. It's a bet I'd love to lose, but I won't.

许多信徒仍在。暗物质猎人正沿着新旧方向拼命奔走，希望自然无法永远保守这个秘密。九月初，作为搜寻已变得何等绝望的一个信号，社区成员因几十个探测器之一中单次无法解释的碰撞的宣布而发狂——那次碰撞甚

至与 WIMP 轮廓不符。我愿赌一千美元：这次潜在的暗物质探测，就像自然在这三十年搜寻中抛给我们的其他每一丝微小希望一样，将会蒸发。这是我乐意输掉的赌注，但我不会输。

In fact, I've begun to wonder where others' optimism comes from and to question one of my own deepest convictions: that the scientific method, given enough time and smart people, will get us over any hurdle, bringing us closer to the truth of things.

事实上，我已开始疑惑他人乐观从何而来，并质疑自己最深的信念之一：科学方法，只要有足够时间与聪明人，就能越过任何障碍，带我们更接近事物的真相。

If dark matter disproves this conviction—if five sixths of all matter will continue to hide for the remainder of human history behind an impenetrable wall of unknowing—then maybe I've been wrong about science itself all along. Maybe all scientists should be grappling with a possibility they've long ignored but which many philosophers have long since accepted: That science will never give us a complete picture of reality. Or, to take this idea to its extreme—as I've begun to do in my darker moments—that our theories are merely a useful but ultimately meaningless facsimile of the world. Dark matter, I've come to believe, could thus become science's greatest lesson.

若暗物质证伪了这一信念——若全部物质的六分之五将在人类历史的剩余时间里继续躲在不可穿透的无知之墙后——那么或许我一直以来对科学本身的看法就是错的。或许所有科学家都应认真对待一种他们长期忽视、而许多哲学家早已接受的可能性：科学永远不会给我们一幅完整的现实图景。或者，把这一想法推到极端——正如我在更阴暗的时刻开始做的——我们的理论只是世

界有用但最终无意义的摹本。我开始相信，暗物质因此可能成为科学最伟大的一课。

The first problem is that dark matter definitely exists. Astronomer Vera C. Rubin spent the 1970s meticulously cataloging the swirl of as many galaxies in the sky as she could. What she found was that the stars near the edge of every galaxy in the observable universe are moving way too fast. Like a racecar hitting a corner with too much speed, these stars should be sent flying off their orbital racetracks and into the void. The gravity of everything we can see at the center of their galaxies simply isn't enough to hold them in.

第一个问题是：暗物质确实存在。天文学家维拉·C·鲁宾在 1970 年代一丝不苟地编录了她所能触及的尽可能多的星系的旋转。她发现，可观测宇宙中每一个星系边缘附近的恒星都移动得太快。就像赛车过弯速度过快，这些恒星本应被甩出轨道赛道，飞入虚空。我们在星系中心所能看见的一切的引力，根本不足以把它们留住。

Rubin's data were airtight. They left two alternatives: either gravity doesn't work the way physicists thought it did, or there's a heap of matter in every galaxy that we can't see.

鲁宾的数据无懈可击。它们留下两个选项：要么引力不像物理学家以为的那样运作，要么每个星系里都有我们看不见的一大堆物质。

The intervening decades have all but killed the former possibility. Astronomers construct gravitational "maps" of galaxy clusters based on tiny deflections of the light passing through those galaxies caused by the matter they contain. Most of this matter exists as a giant, invisible cloud superimposed on the stuff we can see with telescopes.

其后数十年几乎扼杀了前一种可能。天文学家根据穿过星系团的光因其中物质造成的微小偏折，构建星系团的引力“地图”。这些物质的大部分，以叠加在我们用望远镜可见之物上的巨大不可见云的形式存在。

These dark matter maps confirm Rubin's observation: the gravity of these galaxies is about five times stronger than their visible stars and dust would suggest. And when astronomers caught two of these clumps of galaxies colliding, they found a powerful confirmation that it was matter, not gravity, that we had been wrong about.

这些暗物质地图证实了鲁宾的观察：这些星系的引力，大约是其可见恒星与尘埃所暗示的五倍强。而当天文学家捕捉到两团这样的星系碰撞时，他们找到了有力确认：我们弄错的是物质，不是引力。

The cataclysm, called the Bullet Cluster, stripped each clump's regular matter from its dark, superimposed cloud. The gas, stars and galaxies slowed down, roiled into a hot maelstrom. But the two ghostly halos of invisible stuff sailed right past each other, leaving their visible counterparts behind. If the solution to Rubin's mystery is a new law of gravity rather than extra stuff, then what's in those fleeing clouds?

这场被称为子弹星系团的灾变，把每团的常规物质从其暗的、叠加的云中剥离。气体、恒星与星系减速，搅成滚烫的漩涡。但两团幽灵般的不可见晕彼此径直掠过，把可见对应物抛在后面。若鲁宾之谜的解答是新的引力定律而非额外物质，那么那些逃离的云里是什么？

The Bullet Cluster is nigh impossible to explain by modifying Isaac Newton and Albert Einstein's theories of gravity. So are other recent observations, such as galaxies with anomalously slight or severe "racecar" problems—there

can be more or less missing matter in different places, but gravity has to look the same everywhere.

子弹星系团几乎不可能通过修改艾萨克·牛顿与阿尔伯特·爱因斯坦的引力理论来解释。其他近期观测也是如此，例如有异常轻微或严重“赛车”问题的星系——不同地方可以有更多或更少缺失物质，但引力必须处处看起来相同。

Even in our oldest snapshot of the universe's early contents, the so-called cosmic microwave background, dark matter's influence is unmissable. A few hundred thousand years after the big bang, "regular matter"—the 16 species of particles that make up the Standard Model of particle physics—was still a swirling soup, too hot to clump under gravity's pull. But the data reveal that dark matter was already starting to coalesce.

即便在我们关于宇宙早期内容的最古老快照——所谓宇宙微波背景——中，暗物质的影响也无可忽视。大爆炸后几十万年，“常规物质”——构成粒子物理标准模型的16种粒子——仍是翻涌的汤，热得无法在引力牵引下结块。但数据揭示，暗物质已经开始凝聚。

All this evidence led physicists to a simple, coherent, frustrating explanation. Five sixths of the universe's matter is a kind we can't see, yet it built everything we can see, and it's passing through us every moment. But what the heck is it made of?

所有这些证据把物理学家引向一个简单、连贯却令人沮丧的解释。宇宙物质的六分之五是我們看不见的一种，然而它建造了我们所能看见的一切，并每时每刻穿过我们。可它到底由什么构成？

Fortunately, theorists had an answer ready. Their favorite theory of matter—called supersymmetry—already

predicted an as yet undiscovered particle, some 50 times the mass of a proton, that could effortlessly explain every apparent contradiction logged by Rubin and her successors. WIMPs so perfectly addressed this cosmic gap that they became known as the "WIMP miracle." And even more miraculously, we might be able to detect them.

幸运的是，理论家们已有答案待命。他们最爱的物质理论——超对称——已经预言了一种尚未发现的粒子，质量约为质子的 50 倍，能轻松解释鲁宾及其后继者记录的每一个明显矛盾。WIMP 如此完美地填补了这一宇宙缺口，以至于被称为“WIMP 奇迹”。更奇迹的是，我们或许能探测到它们。

WIMPs would interact with regular matter mainly through their collective gravitational pull. But the individual particles also had a minuscule bridge to the Standard Model: they could interact with its particles through the weak force. This meant a WIMP passing through a block of regular matter would usually whiz through like a ghost, but very rarely this ghost would become momentarily corporeal—the WIMP would collide with an atom.

WIMP 主要通过其集体引力牵拉与常规物质相互作用。但单个粒子与标准模型也有一座微小的桥梁：它们可通过弱力与其粒子相互作用。这意味着穿过一块常规物质的 WIMP 通常会像幽灵一样掠过，但极偶尔这幽灵会暂时变得有形——WIMP 会与一个原子碰撞。

In 1985 Witten and Mark Goodman proposed a simple-sounding idea: fill a tank with atoms that flash when struck, put it underground where cosmic rays can't light it up, and wait for dark matter particles to make their identity known. It seemed like the logical next step, the way things had worked throughout the 20th century, when

physicists used similar detectors to pin down each of the Standard Model's characters one by one. Dark matter, physicists figured, would pan out no differently than earlier mysteries, such as how spin works or what protons and neutrons are made of. We'd trap the mysterious entity in a laboratory and then carefully tease out its workings.

1985 年，威滕与马克·古德曼提出了一个听起来简单的想法：用被撞击时会闪光的原子装满一个罐子，把它放在宇宙射线照不到的地下，然后等待暗物质粒子显露身份。这似乎是合乎逻辑的下一步，也是整个 20 世纪事情的运作方式——物理学家用类似探测器一个个钉住标准模型的角色。物理学家们以为，暗物质会与更早的谜题——如自旋如何运作，或质子与中子由什么构成——并无不同。我们会在实验室困住神秘实体，然后仔细梳理其运作。

"I was giving talks, saying, 'We're going find this in 10 years,' and everybody thought that was true," says Katherine Freese, a theoretical astrophysicist at the University of Texas. Freese co-authored a 1986 paper that filled in the details of Goodman and Witten's proposal, establishing "direct" dark matter detection as a field. "We were really convinced, like, 'My God, we can do this,'" she says. "We thought it was right around the corner then."

“我做报告时会说，‘我们十年内就会找到这个’，人人都觉得那是真的，”得克萨斯大学理论天体物理学家凯瑟琳·弗里兹说。弗里兹合著了 1986 年一篇论文，充实了古德曼与威滕提案的细节，确立了“直接”暗物质探测作为一个领域。“我们真的确信，就像‘天哪，我们能做到’，”她说。“那时我们以为就在拐角处。”

I joined the arms race near its peak, almost two decades after Freese and her cadre sparked it. New experiments

were launching every year, battling to cover the full span of what the WIMP's mass and cross section (precisely how ghostly it is) could possibly be.

我在军备竞赛接近高峰时加入，几乎是在弗里兹及其同辈点燃它将近二十年之后。每年都有新实验启动，争相覆盖 WIMP 质量与截面（精确地说，它有多像幽灵）可能的全部范围。

One technology had taken a clear lead: detectors filled with liquid xenon, pioneered by Columbia University physicist Elena Aprile. It's hard to point to a single person more responsible for our tearing through so much of the WIMP's massive parameter space so quickly. "She really is the one who got the use of xenon going," Freese says. So I asked Aprile to be my doctoral adviser in 2015. To me, dark matter was the biggest embarrassment in physics, and I couldn't imagine a higher priority than using the scientific method to fix it.

一种技术已取得明显领先：由哥伦比亚大学物理学家埃莱娜·阿普丽尔开创的液氙探测器。很难说还有哪一个人要对我们如此迅速地撕开 WIMP 巨大参数空间负有更大责任。“真正让氙用起来的人就是她，”弗里兹说。于是我在 2015 年请阿普丽尔做我的博士导师。对我而言，暗物质是物理学最大的尴尬，我无法想象有比用科学方法修好它更高的优先事项。

The first blow to this resolve came in 2018. We had in hand an unprecedented chunk of data from the most sensitive experiment in history, XENON1T. It was poised to probe a great swath of the remaining masses and cross sections the WIMP could still have. Using fake data to blind ourselves from potential bias, we'd built up an elaborate system to automatically analyze the real thing. Now our

entire collaboration had joined a conference call to press a button and watch the results unfold together. We saw our years of work collapse in seconds. I'll never forget the sight of despair washing across a Zoom screen full of faces as our search turned up nothing.

对这一决心的第一次打击来自 2018 年。我们手中握有来自有史以来最灵敏实验 XENON1T 的前所未有的数据块。它蓄势待发，要探测 WIMP 仍可能拥有的剩余质量与截面的一大片。我们用假数据致盲自己以防偏见，建起一套精细系统自动分析真实数据。如今整个合作组加入电话会议，按下按钮，一起观看结果展开。我们看到多年工作在数秒内崩溃。我永远不会忘记，当我们的搜寻一无所获时，绝望冲过 Zoom 屏幕上满屏面孔的景象。

When I went to Italy to help build that experiment's successor, XENONnT, I started polling my peers and soon realized how little of the confidence Freese spoke of remained. That work, too, would go on to reveal nothing, as did its many competitors.

当我去意大利帮助建造该实验的后继者 XENONnT 时，我开始向同行做民调，很快意识到弗里兹所说的那种信心所剩无几。那项工作后来同样一无所获，其众多竞争者亦然。

Physicists are already planning another round of experiments, based on theorists' updated estimates of just how weakly interacting WIMPs can be. But the original miracle they pulled from their desk drawer is dead. These modified WIMPs must be tethered even more loosely to the Standard Model, by way of some additional, unknown particles.

物理学家们已在规划新一轮实验，基于理论家们对

WIMP 究竟能弱相互作用到什么程度的更新估计。但他们从抽屉里抽出的原初奇迹已死。这些修改版 WIMP 必须通过某些额外的未知粒子，与标准模型绑得更松。

But even if this updated guess is right, there's a good chance we'll never find what we're looking for. Our bodies and underground detectors alike are constantly pummeled by billions of neutrinos produced in the sun and atmosphere. Neutrinos interact far more rarely than the original WIMP but similarly to or more often than any version of WIMPs that survived the 2010s. And their interactions are impossible to distinguish from dark matter's. Xenon experiments have already begun to see traces of this impenetrable "neutrino floor."

但即便这一更新猜测正确，我们也很可能永远找不到所寻之物。我们的身体与地下探测器一样，不断被太阳与大气中产生的数十亿中微子轰击。中微子的相互作用远比原初 WIMP 罕见，但与在 2010 年代幸存下来的任何版本 WIMP 相似或更频繁。而其相互作用不可能与暗物质的区分开来。氙实验已开始看到这道不可穿透的“中微子地板”的痕迹。

So the community has moved on to other hypotheticals—to axions, a long-standing alternative hypothesis theorists have had ready-made since 1977, or to lower-mass WIMPs whose tiny bump wouldn't light up a xenon atom. People are paying new attention to the possibility that dark matter is made of primordial black holes, hypothetical entities dating to the big bang that Stephen Hawking predicted in 1971. Each of these black holes swirling around the galaxy would have a mass maybe a bit less than the moon's packed into a microscopic ball.

于是社区转向其他假说——轴子，理论家们自 1977 年

以来就准备好的长期备选；或质量更低的 WIMP，其微小撞击不会点亮氦原子。人们重新关注暗物质由原初黑洞构成的可能性——史蒂芬·霍金 1971 年预言的、可追溯到大爆炸的假想实体。这些在星系中盘旋的黑洞每一个，质量或许略小于月球，却被压进微观的球。

But what if dark matter is none of these things? What if it is not connected to our world through any force besides gravity? In that case, no conceivable technology could sense a single particle of the stuff. Dark matter would be like a shadow realm, superimposed on but disconnected from our own except through its collective tug, its denizens utterly unknowable to us.

可如果暗物质这些都不是呢？如果它除了引力外不通过任何力与我们的世界相连呢？那样的话，没有任何可设想的技术能感知到这种物质的单个粒子。暗物质会像一个影子领域，叠加在我们的世界上却与之脱节，仅通过其集体牵拉相连，其居民对我们完全不可知。

Call me naive, but I never thought this could happen. The scientific method is built to encounter, interrogate and ultimately subsume any mystery it runs across. So I figured dark matter was just another speed bump. Give us enough time, and we'd figure it out like we had so many previous puzzles.

说我天真吧，但我从未想过这会发生。科学方法本是为遭遇、审讯并最终吞没它所碰到的任何谜题而建。所以我以为暗物质只是又一个减速带。给我们足够时间，我们会像解决过去那么多谜题一样把它弄明白。

In retrospect, all those recent wins might have misled me. "Particle physics during the 20th century was incredibly successful," says Simon Allzén, a philosopher at Stockholm University. At blackboards and colliders, my prede-

cessors had conquered and cataloged a zoo of particles into the elegant, orderly Standard Model. They came to expect a "five-sigma" standard of evidence, meaning a true discovery can have only a one-in-several-million chance of being a statistical fluke. Maybe that bar is just too high to maintain. "This has been a golden era," Allzén says. "Modeling our scientific method on what we achieved with the Standard Model is a bad idea."

回顾起来，那些近期胜利或许误导了我。“20 世纪的粒子物理无比成功，”斯德哥尔摩大学哲学家西蒙·阿尔岑说。在黑板与对撞机上，我的前辈们征服并编录了一个粒子动物园，装进优雅有序的标准模型。他们开始期待“五西格玛”证据标准，意味着真正的发现只能有几百万分之一的统计侥幸几率。或许那道门槛高得难以维持。“这是一个黄金时代，”阿尔岑说。“以我们在标准模型上取得的成就来塑造科学方法，是个坏主意。”

Stop exalting physics above the other sciences—after all, as Allzén puts it, we believe in dinosaurs even though we can't trap one in a lab—and maybe dark matter isn't such a problem. "We have detected it," argues Siska De Baeremaeker, a philosopher at Stockholm University. "That you haven't found something until you've found it in a lab is weirdly dismissive of gravitational detection."

停止把物理学抬高到其他科学之上——毕竟，正如阿尔岑所说，我们相信恐龙，即使无法在实验室困住一只——或许暗物质就不是那么大的问题。“我们已经探测到它了，”斯德哥尔摩大学哲学家西斯卡·德·巴尔德马克尔争辩道。“非要在实验室找到才算找到，是对引力探测的怪异轻蔑。”

Dark matter is there. Using just its gravity, we've built extensive catalogs of the shadowy stuff's presence and evo-

lution across the eons. Maybe we should accept that the secondary goal—figuring out what its constituents are—might be beyond reach. "That would be the worst-case scenario for physics, especially for particle physics," De Baerdemaeker says. "But nature doesn't have to play nice."

暗物质就在那里。仅用其引力，我们就已建立了关于这阴影物质跨越亿万年的存在与演化的详尽目录。或许我们应接受：次要目标——弄清其组分是什么——或许超出了触及范围。“那对物理学、尤其是粒子物理会是最坏的情况，”德·巴尔德马克尔说。“但自然不必友善。”

Still, I can't quite shake my training. Atoms are made up of electrons and nucleons, which consist of up and down quarks. How can we call science a success if we'll never know what this other stuff—which is almost all of the stuff—is made of?

尽管如此，我仍甩不掉所受的训练。原子由电子与核子构成，核子由上下夸克组成。若我们永远不知道这另一种东西——它几乎是全部东西——由什么构成，我们怎能称科学为成功？

Dark matter has me thinking I've been wrong about what science is even for. I thought it was the ultimate place for asking humankind's biggest questions (How did we get here? What is everything made of? How does it all come together?) because it was the only place where you could get answers that were simply true. But are they?

暗物质让我开始觉得，自己一直误解了科学究竟是为了什么。我曾以为它是追问人类最大问题（我们如何来到这里？一切由什么构成？这一切如何合在一起？）的终极场所，因为它是唯一能得到单纯为真之答案的地方。可它们是吗？

Every theory we've ever had has technically been wrong—an imperfect approximation of reality—and we can be sure the Standard Model is no different. Dark matter is just the most glaring of its gaps, different only in degree from its other fuzzy impressions of the world, from the electron to the Higgs boson.

我们曾有过的每一个理论在技术上都是错的——对现实的不完美近似——我们可以肯定标准模型也不例外。暗物质只是其缺口中最为刺眼的一个，与它对世界其他模糊印象——从电子到希格斯玻色子——只有程度上的不同。

I'm starting to think none of these objects are any more "real" than the WIMP. They're just theoretical ideas that help us predict what will happen, like how high school-level mechanics can predict how far away a punted football will land. I'm starting to think that is science's actual job: predicting what will happen. The Standard Model is just a fancier version of those high school equations, a reliable framework for saying what will come out when you smash matter together at high speeds. Maybe the search for absolute truth—not just a reliably predictive picture of the world but a meaningful one—was always too ambitious.

我开始觉得，这些对象并不比 WIMP 更“真实”。它们只是帮助我们预测将发生什么的理论观念，就像高中级力学能预测被踢出的橄榄球会落在多远。我开始觉得那才是科学的实际工作：预测将发生什么。标准模型只是那些高中方程的更花哨版本，一个可靠框架，用于说明当你以高速把物质撞在一起时会出来什么。或许对绝对真理的追寻——不只是一幅可可靠预测的世界图景，而是一幅有意义的图景——从来都过于野心勃勃。

Tim Maudlin, a philosopher of science at New York University, says that I've retreated too far, that I should "lighten up a little." Our theory may never be completely finished, he points out, but that doesn't mean entities such as the electron have zero connection to reality. But Maudlin also warns against my former unassailable optimism, calling it a kind of "theological thesis." There may be fundamental or practical limitations that stop us from ever answering a scientific question. "I mean, that's life, buddy," he says. "The world doesn't care about you."

纽约大学科学哲学家蒂姆·莫德林说我退得太远了，我应该“放松一点”。他指出，我们的理论或许永远不会完全完成，但这并不意味着像电子这样的实体与现实零连接。但莫德林也警告反对我从前那种不可动摇的乐观，称之为一种“神学论题”。可能存在根本或实践上的限制，阻止我们永远回答某个科学问题。“我是说，伙计，生活就是这样，”他说。“世界不在乎你。”

Something like faith might be what it actually takes to be a scientist, even if it isn't totally rational. In fact, most of the physicists I've talked to still believe we'll make sense of dark matter someday. "I do think we'll get it," Freese says. "Just because we haven't solved a problem doesn't mean the next generation won't be smart enough to figure it out." Aprile, my former adviser, thinks the right stroke of experimental genius is all that's lacking. "We have to invent a tool. The progress has always only been made because Galileo invented the telescope," she says. "Some smart kid one day will figure it out. I know it must be."

某种类似信仰的东西，或许才是成为科学家真正所需，即便它并非完全理性。事实上，我交谈过的多数物理学家仍相信我们有朝一日会弄清暗物质。“我确实认为我们会搞明白，”弗里兹说。“仅仅因为我们还没解决一个问

题，并不意味着下一代不够聪明到把它弄清楚。”我的前导师阿普丽尔认为，所缺的只是正确的那一记实验天才的一击。“我们必须发明一种工具。进步之所以发生，从来只是因为伽利略发明了望远镜，”她说。“哪天某个聪明孩子会搞明白。我知道一定是这样。”

Scientists have never really been motivated by a desire to predict where the football will land. They're dreamers in search of answers, of meaning, whether it's there or not, even if they have to strive toward a receding horizon. "What else is there—give up? And then what? Do something else?" Aprile asks me these questions rhetorically. Her path was never a choice. "I'm never happy, actually—that's the problem," she says. "Ever since I was young, I wanted to get out, I wanted to do more."

科学家们从未真正被预测橄榄球落点的欲望所驱动。他们是追寻答案、追寻意义的梦想者，无论那意义是否存在，即便他们必须奔向一道不断后退的地平线。“还能怎样——放弃？然后呢？做点别的？”阿普丽尔修辞性地问我这些问题。她的道路从来不是选择。“实际上我从不快乐——那就是问题，”她说。“从年轻时起，我就想出去，我想做得更多。”

Someone once told me that the most beautiful thing about science is how it points at its own limitations. Maybe that's the beauty of dark matter. Maybe Aprile and the other restless dreamers have dug up the ultimate limit—one the rest of us can't ignore.

有人曾告诉我，科学最美的地方在于它如何指向自身的局限。或许那就是暗物质之美。或许阿普丽尔与其他不安分的梦想者挖出了终极极限——我们其余人无法忽视的极限。

The hunt for reality's smallest pieces

追寻实在最小的碎片

The hunt for reality's smallest pieces

By Jay Bennett · September 15, 2026 · edited by Clara Moskowitz

<https://www.scientificamerican.com/article/what-are-the-smallest-pieces-of-reality/>

Physicists are searching for what lies underneath the particles we know

物理学家们正在搜寻我们已知粒子之下还有什么

On June 25, 2012, particle physicist Sau Lan Wu was in Building 32 at CERN, the European particle physics laboratory near Geneva. Hundreds of feet below, the Large Hadron Collider (LHC)—the world's biggest scientific instrument—had been conducting high-energy particle collisions for more than two years. The machine accelerates protons to near the speed of light and slams them into one another to produce sprays of fundamental particles and radiation. "Suddenly there's a lot of noise," Wu recalls. "We heard one of my students screaming."

2012 年 6 月 25 日，粒子物理学家吴秀兰正在日内瓦附近欧洲粒子物理实验室 CERN 的 32 号楼。数百英尺之下，世界最大的科学仪器——大型强子对撞机（LHC）——已进行了两年多的高能粒子碰撞。这台机器把质子

加速到接近光速，再让它们彼此撞击，产生基本粒子与辐射的喷涌。“突然一阵嘈杂，”吴回忆道。“我们听见我的一个学生在尖叫。”

Wu had assigned one of her most capable statisticians, graduate student Haoshuang Ji, to the task of confirming whether the long-sought Higgs boson particle had briefly appeared among the collider's showers of subatomic particles. Ji had worked through the night to combine two datasets tracking different potential signs of the Higgs.

吴把确认对撞机亚原子粒子雨中是否短暂出现了长期寻找的希格斯玻色子的任务，交给了她最能干的统计学家之一、研究生纪浩双。纪通宵工作，合并了追踪希格斯不同潜在迹象的两套数据集。

The combined statistical significance of the datasets surpassed the threshold needed to claim a discovery. The Higgs boson was real. Celebratory cheers erupted down the corridor. "Everybody smiled more and more; our mouths became bigger and bigger," Wu says. "This is what you call the joy of discovery."

合并后的统计显著性超过了宣布发现所需的阈值。希格斯玻色子是真实的。走廊里爆发出庆祝的欢呼。“人人笑得越来越开；我们的嘴越张越大，”吴说。“这就是所谓发现的喜悦。”

The Higgs boson, announced to the world nine days later, brought the number of known elementary particles in the universe to 17. It was the last particle predicted by the Standard Model, theorists' best description of nature's fundamental bits. The model groups particles into fermions (the building blocks of matter, such as quarks) and bosons (force-carrying particles such as photons).

九天后向世界宣布的希格斯玻色子，把宇宙中已知基本粒子的数量带到了 17。它是标准模型——理论家对自然基本构件的最佳描述——预言的最后一粒粒子。该模型把粒子分为费米子（物质的构件，如夸克）与玻色子（携带力的粒子，如光子）。

These particles are the smallest bits of nature known. But can they be broken down further? "That's what we search for in experiments," says James Beacham, a physicist at CERN. "There's this notion of what it means to be the smallest possible thing. Really we're asking: What are the most fundamental ways by which the universe organizes itself? What is physical reality? It's really a kind of profound, almost philosophical question."

这些粒子是已知自然最小的碎片。但它们能否被进一步拆解？“那正是我们在实验中搜寻的，” CERN 物理学家詹姆斯·比彻姆说。“存在着‘最小可能之物’意味着什么这一观念。我们真正在问的是：宇宙组织自身的最根本方式是什么？物理实在是什么？这真是一种深刻的、近乎哲学的问题。”

With more time and more advanced tools, scientists wonder, will we keep discovering more underlying pieces of the universe? Will we ever reach the limit—the final building block of nature? How can we truly know if we've drilled down as far as the subatomic layers go?

有了更多时间与更先进的工具，科学家们想知道：我们是否会不断发现宇宙更多底层碎片？我们是否会抵达极限——自然最终的建筑块？我们如何真正知道，我们已经钻到了亚原子层次所能到达的尽头？

In addition to looking for smaller ingredients within known particles, physicists believe that there could be additional elementary particles that have yet to be identified. A

hypothetical dark matter particle, for example, may be composed of smaller parts that we can think of as "dark quarks," Beacham says.

除了在已知粒子中寻找更小的成分，物理学家们相信还可能存在尚未鉴定的额外基本粒子。例如，一种假想的暗物质粒子，或许由我们可以想成“暗夸克”的更小部分构成，比彻姆说。

The search for new fragments of reality may reveal that our understanding of elementary physics is flawed. The Standard Model represents particles as points, meaning they have no volume. Scientists, however, know that they don't literally have zero volume, but they are so small that there is no mathematical reason to include their size in physical models. (And according to the model, particles are merely "excitations" in fundamental fields that pervade the universe.)

对新实在碎片的搜寻，可能揭示我们对基本物理的理解有缺陷。标准模型把粒子表示为点，意味着它们没有体积。然而科学家们知道它们并非字面意义上体积为零，只是如此之小，以至于没有数学理由在物理模型中纳入其尺寸。（而按该模型，粒子不过是遍布宇宙的基本场中的“激发”。）

But what if instead of points, these bits of mass and energy are actually something else, like tiny, vibrating strings? Under string theory, a particle's characteristics would then be determined by the string's properties and the way it mingles with other strings, all quivering and coiling near the infinitesimal limits of physical size.

可如果这些质量与能量的碎片不是点，而是别的东西——比如微小的振动弦呢？在弦论下，粒子的特性将由

弦的性质及其与其他弦的交织方式决定，所有这些弦都在物理尺度的无穷小极限附近颤动、盘绕。

“Think of strings interacting,” says Mirjam Cvetič, a theoretical physicist specializing in string theory at the University of Pennsylvania. “Open strings can combine and split. When they combine, they can make closed strings.” Most elementary particles would be formed by open strings, she says, and closed strings would make up the hypothetical particles that carry the gravitational force, called gravitons.

“想想弦的相互作用，”宾夕法尼亚大学专攻弦论的理论物理学家米尔亚姆·茨韦蒂奇说。“开弦可以结合与分裂。当它们结合时，可以形成闭弦。”她说，多数基本粒子由开弦形成，而闭弦构成携带引力的假想粒子——引力子。

To accurately explain known physics, these strings must exist across at least 10 dimensions, a geometric structure that is assumed to be accessible only at tiny scales. Opening, closing and thrumming like guitar chords across layers of reality, the strings of the theory are capable of explaining the established laws of physics in literally a quadrillion different ways. To critics, this makes the theory untestable—even worthless. But to proponents, the idea has a certain kind of beauty in its ability to explain facets of the universe. That’s “not a proof,” Cvetič says, “but a compelling feature.”

要准确解释已知物理，这些弦必须存在于至少 10 个维度中，这一几何结构被假定仅在微小尺度上可及。弦论中的弦像吉他和弦一样开合、嗡鸣，跨越实在的层次，能够以字面上一千万亿种不同方式解释既定物理定律。对批评者而言，这使该理论不可检验——甚至毫无价值。但

对支持者来说，这一想法在解释宇宙诸面相的能力上有某种美。那“不是证明，”茨韦蒂奇说，“却是一个引人注目的特征。”

An alternative idea to string theory is the proposal that spacetime itself is made of tiny loops. Known as loop quantum gravity, this theory proposes that space isn't smooth; instead it can be broken down into discrete bits.

弦论的一个替代想法是：时空本身由微小的环构成。这一被称为圈量子引力的理论提出，空间并不平滑；相反，它可被分解为离散的比特。

Under loop quantum gravity, spacetime is a "sea of gravity that is in a tempest," says Francesca Vidotto, a theoretical physicist at Western University in Ontario. Woven together, these fundamental loops of gravity can mathematically account for the curvature of spacetime.

在圈量子引力下，时空是“暴风雨中的引力之海，”安大略省西安大略大学理论物理学家弗朗西斯卡·维多托说。编织在一起，这些基本的引力环可以在数学上解释时空的曲率。

Vidotto says this scenario is similar to being bathed in light, which is described through the electromagnetic force. "We are immersed in the electromagnetic field," she says. "And when we try to interact at a quantum level with this electromagnetic field, the only interactions we can have are through these discrete objects, the photons." The same relation may exist for gravity. "I think for spacetime we should have a similar intuition that we are immersed in the gravitational field," she says. "And when we try to interact with this field at the smallest possible scale, we discover that there is this quantum spacetime, like grains."

维多托说，这一情景类似于沐浴在光中，而光通过电磁力来描述。“我们沉浸在电磁场中，”她说。“当我们试图在量子层面上与这个电磁场相互作用时，我们唯一能有的相互作用是通过这些离散对象——光子。”对引力或许存在同样的关系。“我认为对时空我们应有类似的直觉：我们沉浸在引力场中，”她说。“而当我们试图在尽可能小的尺度上与这个场相互作用时，我们发现存在这种量子时空，像谷粒一样。”

One tantalizing prediction of loop quantum gravity is that tiny remnants of black holes could coexist with so-called white holes—regions of spacetime that are essentially the opposite of black holes and prohibit light and matter from entering. The two could combine through quantum superposition (the ability of quantum objects to occupy multiple states at once) to form stable particles. These particles are a theoretical candidate for dark matter. "If we are right," Vidotto says, "it's as if we had quantum gravity under our nose all these years without realizing."

圈量子引力的一个诱人预言是：黑洞的微小残余物可能与所谓白洞共存——白洞本质上是黑洞的对立面，禁止光与物质进入。两者可通过量子叠加（量子对象同时占据多个态的能力）结合，形成稳定粒子。这些粒子是暗物质的一个理论候选者。“如果我们对的，”维多托说，“那就仿佛这些年来量子引力一直在我们鼻子底下，而我们却没有意识到。”

If strings or quantum loops exist, physicists believe they would be about the size of the Planck length, which is thought to represent a physical limit to how small something can be, about 1.6×10^{-35} meter. According to quantum theory, it is impossible to measure beyond this limit because anything smaller would have physical uncertain-

ties that would prevent it from having a definite size.

若弦或量子环存在，物理学家们相信它们大约是普朗克长度的大小，后者被认为代表某物能有多小的物理极限，约 1.6×10^{-35} 米。根据量子理论，不可能测量超出这一极限，因为任何更小的东西都会有物理不确定性，阻止它具有确定的尺寸。

To experiment close to the Planck scale, we would need a circular particle collider centered on the sun and reaching "a little past the outer orbit of Neptune," Beacham says. This grand space accelerator won't be possible anytime soon, but scientists are working on upgrades at the Large Hadron Collider that are slated to come online in 2030. These modifications will allow the machine to measure even more particles erupting from collisions. The primary objective is to learn more about the Higgs boson, which may actually be multiple types of particles or could be made of composite parts.

要在接近普朗克尺度处做实验，我们需要一座以太阳为中心、延伸到“略超过海王星外轨道”的环形粒子对撞机，比彻姆说。这座宏伟的太空加速器短期内不可能建成，但科学家们正在为计划于 2030 年上线的大型强子对撞机升级而工作。这些改造将使机器能测量碰撞中喷涌出的更多粒子。主要目标是更多地了解希格斯玻色子——它实际上可能是多种粒子，或可能由复合部分构成。

"What are the subtle deviations of how this Higgs boson behaves?" wonders Archana Sharma, a physicist at CERN who helped to design, build and operate one of the LHC's primary detectors. "How does it decay? How does it self-interact? Are there two bosons that can be produced or not?"

“这个希格斯玻色子行为的微妙偏差是什么？”曾帮助设

计、建造并运行 LHC 主要探测器之一的 CERN 物理学家阿尔查娜·夏尔马想知道。“它如何衰变？它如何自相互作用？能否产生两个玻色子？”

Because the Higgs has a quantum spin of zero, it is the only known elementary scalar particle. Learning more about how it behaves is crucial to understanding the ghostly medium responsible for imparting mass in the universe. "The Higgs boson could be the doorway to an entirely new physics," Sharma says. "It's unlike any other particle we know."

因为希格斯的量子自旋为零，它是唯一已知的基本标量粒子。更多地了解它如何行为，对理解赋予宇宙质量的幽灵媒介至关重要。“希格斯玻色子可能是通向全新物理的大门，”夏尔马说。“它不像我们知道的任何其他粒子。”

With any luck, continuing to up the ante in the LHC could reveal new facets of reality. Whether we will ever fully understand the underpinnings of existence, however, is anyone's bet. "Some questions may be unanswerable. That's, of course, very, very important to accept," Sharma says. "And we need to continue to ask nevertheless."

运气好的话，继续在 LHC 加码可能揭示实在的新侧面。然而我们是否终将完全理解存在的根基，则是任何人的赌注。“有些问题或许无法回答。当然，接受这一点非常重要，”夏尔马说。“而我们仍然需要继续追问。”

What if 'consciousness' isn't real?

如果「意识」并不真实？

What if 'consciousness' isn't real?

By Tim Bayne · September 15, 2026 · edited by Allison Parshall

<https://www.scientificamerican.com/article/what-if-consciousness-isnt-real/>

Our understanding of consciousness could be fundamentally wrong

我们对意识的理解可能从根本上就是错的

I write this from Santiago, nestled between the snow-capped Andes to the east and the Chilean Coastal Range to the west. I am here with about 500 other consciousness researchers for our most important annual meeting. A key theme of this year's gathering is the distribution of consciousness: When does consciousness first emerge in human development? Which animals are conscious, and might it occur in artificial-intelligence systems? And can we construct a consciousness detector, something that might tell us whether, say, babies, bees or bots are members of the "consciousness club"?

我在圣地亚哥写下这些文字，东边是白雪皑皑的安第斯山脉，西边是智利海岸山脉。我与大约 500 名其他意识研究者一起，参加我们最重要的年度会议。今年聚会的

一个关键主题是意识的分布：意识在人类发育中何时首次出现？哪些动物有意识，它是否可能出现在人工智能系统中？我们能否建造一台意识探测器，告诉我们婴儿、蜜蜂或机器人是否属于“意识俱乐部”的成员？

These questions are among the most intractable of scientific problems. Consciousness researchers like me have been tackling them for decades. We've made some progress, yet agreement about how to tell whether an organism or machine is conscious remains elusive. Perhaps part of the problem lies with the question itself. Perhaps "consciousness" as a term doesn't really belong to the language of science.

这些问题属于最棘手的科学难题之列。像我这样的意识研究者已与它们搏斗数十年。我们取得了一些进展，但关于如何判断一个有机体或机器是否有意识，共识仍然难以捉摸。或许问题的一部分出在问题本身。“意识”作为一个术语，或许并不真正属于科学语言。

Consciousness researchers have never agreed on what we mean when we use that word. In some disciplines, "consciousness" means "sentience"; in others, "awareness." Still others use it interchangeably with "experience" or "qualia." The muddy terminology makes it hard to tell whether the different areas of consciousness even have a common subject matter.

意识研究者们从未就我们使用这个词时的含义达成一致。在某些学科中，“意识”意味着“感受性”；在另一些学科中，意味着“觉知”。还有人把它与“经验”或“感受质”互换使用。术语泥泞，使人很难判断意识的不同领域是否甚至有共同的主题。

Consciousness hasn't always been central to discussions of the mind. Perhaps it shouldn't be.

意识并非总是心灵讨论的中心。或许它本不该是。

If appealing to synonyms doesn't work, maybe a better strategy for saying what we mean by "consciousness" is to point to clear, uncontroversial examples of it. In the same way that you might explain what "jazz" means by pointing to the music of Miles Davis or Ella Fitzgerald, so, too, might you explain what "consciousness" means by pointing to familiar experiences—feeling sand between your toes, tasting a ripe strawberry, seeing the sun reflect off a snow-capped mountain peak.

若诉诸同义词行不通，或许更好的策略是指出清晰、无争议的例子，来说明我们所谓“意识”是什么。正如你可以通过指向迈尔斯·戴维斯或埃拉·菲茨杰拉德的音乐来解释“爵士”的含义，你也可以通过指向熟悉的经验——脚趾间的沙子、品尝成熟草莓、看见阳光反射在白雪皑皑的山峰上——来解释“意识”的含义。

These are all states that it "feels like something" to be in. In contrast, there is nothing that it feels like to be dreamlessly asleep or deeply sedated. In the 1990s philosopher Ned Block, then at the Massachusetts Institute of Technology, introduced the term "phenomenal consciousness" to refer to this state. It's a simple concept that has played a central role in the past three decades of consciousness science and guided discussions about who and what is conscious. Debates about whether it "feels like something" to be a baby, bee or bot are widely thought to carry ethical significance, marking the contrast between entities that have intrinsic moral status and those that don't.

这些都是“感觉起来像某种东西”的状态。相比之下，无梦沉睡或深度镇静时，没有任何感觉。1990 年代，当时在麻省理工学院的哲学家内德·布洛克引入了“现象意识”一词来指称这一状态。这是一个简单概念，在过去三十年的意识科学中扮演了核心角色，并引导了关于谁与什么是有意识的讨论。关于婴儿、蜜蜂或机器人是否“感觉起来像某种东西”的辩论，被广泛认为具有伦理意义，划出了拥有内在道德地位的实体与没有的实体之间的对比。

But despite its apparent universality, phenomenal consciousness has not always occupied center stage in how human beings have approached the mind. In fact, it is far from a universal concept. In the wonderfully named 1988 paper "Yishi, Duh, Um, and Consciousness," University of Oxford philosopher Kathleen V. Wilkes argued that there are no synonyms for "phenomenal consciousness" in ancient Greek, Mandarin, Croatian or even pre-17th-century English. (The "yishi" of Wilkes's title is perhaps the closest that Mandarin comes to a term for consciousness; "duh" and "um" are Croatian terms that have been offered as possible translations for "mind.")

但尽管看似普遍，现象意识并非总是占据人类对待心灵方式的中心舞台。事实上，它远非一个普遍概念。在那篇美妙地命名为《意识、Duh、Um 与 Consciousness》的 1988 年论文中，牛津大学哲学家凯瑟琳·V·威尔克斯论证道：古希腊语、普通话、克罗地亚语乃至 17 世纪以前的英语中，都没有“现象意识”的同义词。（威尔克斯标题中的“yishi”或许是普通话最接近意识的词；“duh”与“um”是被提出作为“心灵”可能译法的克罗地亚语词。）

Here in the Santiago region, the Indigenous language of

Mapudungun addresses questions of mentality using the term "rakizuam." This word captures some of what's meant by "consciousness," but it includes reflexive awareness of interdependence with other people and the natural world—relational elements that "consciousness" fails to include. One might wonder what debates about other minds might have looked like had Mapudungun, rather than English, established itself as the language of science.

在圣地亚哥地区，原住民语言马普切语用“rakizuam”一词处理心智问题。这个词捕捉了“意识”的部分含义，但它包含对与他人及自然界相互依存的反思性觉知——“意识”未能纳入的关系性要素。人们或许会想，若是马普切语而非英语成为科学语言，关于他心的辩论会是什么样子。

The dearth of synonyms for "phenomenal consciousness" settles nothing in and of itself. (None of these languages contains synonyms for "quarks" or "apoptosis" either.) But it does challenge the natural assumption that understanding the space of possible minds requires appealing to the concept of consciousness. Consciousness—or at least our concept of consciousness—hasn't always been central to discussions of the mind. Perhaps it shouldn't be central.

“现象意识”同义词的匮乏本身并不能解决任何问题。（这些语言中也没有“夸克”或“细胞凋亡”的同义词。）但它确实挑战了一种自然假定：理解可能心灵的空间必须诉诸意识概念。意识——或至少我们关于意识的概念——并非总是心灵讨论的中心。或许它本不该是中心。

Science requires concepts that support prediction, explanation and understanding. A good scientific concept is one that cuts nature "at its joints," bundling together

phenomena that share an underlying nature. The English term "consciousness" may turn out to play that role. But it's also possible that "consciousness" will fail to make the cut—because it either bundles together phenomena that don't really belong together or fails to bundle together things that do belong together (or both).

科学需要支持预测、解释与理解的概念。一个好的科学概念是在自然的“关节”处切开，把共享底层本质的现象捆在一起。英语词“consciousness”或许最终能扮演那个角色。但也有可能“意识”通不过筛选——因为它要么把并不真正属于一起的现象捆在一起，要么未能把确实属于一起的东西捆在一起（或两者皆有）。

Precedents from this history of science are not hard to find. As philosopher Patricia Churchland once observed, the ancients grouped a wide variety of disparate phenomena together as forms of "fire": combustion (the burning of wood); fusion (activity on the sun); electrically induced incandescence (lightning); bioluminescence (the activity of fireflies); and spectral emission (the Northern Lights). Galileo reshaped our conception of velocity by distinguishing average velocity from instantaneous velocity, and 18th-century physicist Joseph Black reshaped our understanding of heat, showing that we needed to distinguish it from temperature.

科学史上的先例不难找到。正如哲学家帕特丽夏·丘奇兰德曾观察到的，古人把各种各样迥异的现象归为“火”的形式：燃烧（木头燃烧）；聚变（太阳上的活动）；电致发光（闪电）；生物发光（萤火虫的活动）；以及光谱发射（北极光）。伽利略通过区分平均速度与瞬时速度重塑了我们对速度的概念，而 18 世纪物理学家约瑟夫·布莱克重塑了我们对热的理解，表明我们需要把它与温度区分开来。

Concepts are tools. And as a tool for describing the structure of our own minds, "consciousness" is extremely useful—perhaps even indispensable. But to understand systems that are radically different from us, we may need to forge new concepts. As AI becomes increasingly sophisticated, we might find it irresistible to apply the language of consciousness to such systems, just as we are often tempted to use the language of consciousness in describing bees or newborn babies. But we need to recognize the limitations of consciousness talk. Certainly we should ask whether bees, babies or bots are conscious. But in asking that question, we should be alive to the possibility that the question itself might need replacing.

概念是工具。而作为描述我们自身心灵结构的工具，“意识”极为有用——或许甚至不可或缺。但要理解与我们截然不同的系统，我们或许需要锻造新概念。随着人工智能日益精密，我们或许会发现把意识语言应用于此类系统难以抗拒，正如我们常被诱惑用意识语言描述蜜蜂或新生婴儿。但我们需要认识意识话语的局限。当然，我们应当追问蜜蜂、婴儿或机器人是否有意识。但在提出那个问题时，我们应当意识到：问题本身或许需要被替换。

What came before the big bang?

大爆炸之前是什么？

What came before the big bang?

By Sarah Scoles · February 10, 2026 · edited by Clara Moskowitz

<https://www.scientificamerican.com/article/what-came-before-the-big-bang/>

Physicists cannot access anything that existed before the start of time and space, but they have theories

物理学家无法触及时间与空间开始之前存在的任何事物，但他们有理论

The big bang wasn't a bang in the traditional sense, but it was the start of important things. Space, for one. Time, for another. And it began the conditions and processes that eventually resulted in us humans, who can sit here and wonder about space and time. The big bang was, effectively, the beginning of the universe. According to human logic, there must have been something before the big bang, even if "before" is the wrong word because there was no time until after.

大爆炸并不是传统意义上的爆炸，但它是重要事物的开端。其一是空间。其二是时间。它也开启了最终造就我们这些人类的条件与过程——使我们能坐在这里思索空间与时间。大爆炸实际上是宇宙的开端。按人类逻辑，大

爆炸之前必定有什么，即便“之前”是个错误的词，因为直到之后才有时间。

The good news is that physicists do have ways of thinking about—and even empirically studying—the origins of the origin of the universe. Counterintuitive and impossible as it may seem, cosmologists are even making progress in determining which wild ideas might peel back the veil on that early era, even though it remains inaccessible to us.

好消息是：物理学家们确实有思考——甚至经验研究——宇宙起源之起源的方法。看似反直觉且不可能，宇宙学家们甚至在推进判定哪些狂野想法可能揭开那一早期时代的面纱，尽管它对我们仍不可及。

For millennia what happened before and at the beginning of the universe was not a question scientists could even scratch at. Cosmological queries were the dominion of philosophers, says Jenann Ismael—herself a philosopher of physics at Johns Hopkins University. The most fundamental query, of course, is where we come from, a question as popular among philosophers as it is with the rest of us. Other questions, Ismael says, include doozies such as “What are space and time? Does time have a beginning? Does space have boundaries?”

数千年来，宇宙开始之前与开始之时发生了什么，甚至不是科学家们能抓挠的问题。宇宙学疑问属于哲学家的领地，约翰斯·霍普金斯大学物理哲学家杰南·伊斯梅尔说——她本人就是。最根本的疑问当然是我们从何处来，这个问题在哲学家中与在其余人中一样流行。其他问题，伊斯梅尔说，包括诸如“空间与时间是什么？时间有开端吗？空间有边界吗？”这类棘手之物。

Even after cosmology became a hard science, the field was a bit sketchy, Ismael says. “The science was one-

and-a-half facts,” she adds. The sentiment, she says, is usually attributed to 20th-century British physicist James Jeans. But that has changed in the past century or so as the philosophers’ musings have wandered into the realm of theory, experiment and data. “These old conceptual questions are arising in ways that have new angles, a new spin and a new framework,” Ismael continues.

即便在宇宙学成为硬科学之后，这个领域也有点粗糙，伊斯梅尔说。“科学那时只有一个半事实，”她补充道。她说，这一说法通常归于 20 世纪英国物理学家詹姆斯·金斯。但在过去大约一个世纪里，随着哲学家的沉思走入理论、实验与数据的领域，情况已经改变。“这些古老的概念问题正以具有新角度、新转向与新框架的方式出现，”伊斯梅尔继续说。

It’s unclear whether science as a discipline, and scientists as people, will ever be able to answer some questions definitively. After all, no one can “see” what happened before the big bang, and no one will ever be able to, at least not directly. But the current and future universe, researchers are learning, may contain clues about the distant past, ways to test theories about the before before. “I’m happy to listen to any framework, but I only start taking it seriously when it produces a clean observational target that a real instrument can go after,” says Brian Keating, a cosmologist at the University of California, San Diego. “If there isn’t a discriminant you can measure, you’re doing metaphysics with equations.”

科学作为一门学科、科学家作为人，是否终能明确回答某些问题，尚不清楚。毕竟，没人能“看见”大爆炸之前发生了什么，也永远没人能——至少不能直接看见。但研究者们正在认识到，当前与未来的宇宙或许包含着关于遥远过去的线索，检验关于“之前的之前”之理论的

途径。“我乐意倾听任何框架，但我只有在它产出一台真实仪器可以追击的干净观测靶标时，才开始认真对待，”加州大学圣地亚哥分校宇宙学家布赖恩·基廷说。“如果没有你可以测量的判别量，你就是在用方程做形而上学。”

Here are three ideas that he and other scientists take seriously about the cosmos's ultimate origins:

以下是他与其他科学家认真对待的关于宇宙终极起源的三个想法：

The No-Boundary Proposal

无边界提议

Quantum mechanics is the physics of the extremely small, ruled by statistics and uncertainty. It's also what may have shaped the early universe. To understand the quantum cosmos, scientists calculate the probability of a given output from a certain input.

量子力学是极小尺度的物理，由统计与不确定性统治。它也可能塑造了早期宇宙。为理解量子宇宙，科学家计算给定输入产生给定输出的概率。

In cosmology, the "output" is the universe as it looks today. "The question is: What should the input be?" says Jean-Luc Lehnert, former head of the Theoretical Cosmology group at the Max Planck Institute for Gravitational Physics (Albert Einstein Institute) in Germany.

在宇宙学中，“输出”是今天看起来的宇宙。“问题是：输入应该是什么？”德国马克斯·普朗克引力物理研究所（阿尔伯特·爱因斯坦研究所）理论宇宙学组前负责人让-吕克·莱纳斯说。

Physicists can break up the problem into chunks of outputs and inputs. If they consider the modern universe to be the output, they can try to figure out what input might have produced it. Then they can step backward by taking that input as a new output and determine what conditions earlier in the universe might have produced that state, and so on. They can theoretically (if they have a lot of time on their hands) do that forever, going in steps to reach the before before—and even before that.

物理学家可以把问题拆成输出与输入的块。若他们把现代宇宙视为输出，他们可以试图弄清什么输入可能产生了它。然后他们可以把那个输入当作新输出，后退一步，判定宇宙更早时什么条件可能产生了那个状态，如此类推。他们理论上（若有大量时间）可以永远这样做，一步步到达“之前的之前”——甚至更早。

That infinite regression, however, didn't make sense to physicists Stephen Hawking and James Hartle, who worked on the question together in the 1980s. They decided to eliminate the universe's ultimate input—its "beginning." Instead they formed a model of the universe called the no-boundary proposal. They suggested that time and space form a closed, rounded surface: a four-dimensional hemisphere of spacetime.

然而，这种无限倒退对 1980 年代一起研究该问题的物理学家史蒂芬·霍金与詹姆斯·哈特尔没有意义。他们决定消除宇宙的终极输入——它的“开端”。相反，他们形成了称为无边界提议的宇宙模型。他们提出，时间与空间形成一个封闭、圆润的表面：时空的四维半球。

Imagine the universe like the globe of Earth. The big bang is the North Pole. There is no "before" it, just as there is no north of north. "Before" becomes irrelevant

as a concept. "It's almost like a Zen idea," Lehnars says. And it's one he's toying with in calculations to see if he can re-create the universe we see today from a round place with no north of north.

把宇宙想象成地球仪。大爆炸是北极。没有“在它之前”，正如没有比北更北的地方。“之前”作为一个概念变得无关。“这几乎像一个禅宗想法，”莱纳斯说。而他正用计算把玩它，看自己能否从一个没有比北更北的圆形地方，重现我们今天所见的宇宙。

"The no-boundary proposal has a decent amount of support, or at least interest, within the physics community," says Sean Carroll, a professor of natural philosophy at Johns Hopkins. He notes that some scientists worry about how well defined the idea is, but he finds it to be a "natural starting point," given what we know about quantum gravity.

“无边界提议在物理学界有相当程度的支持，或至少是兴趣，”约翰斯·霍普金斯大学自然哲学教授肖恩·卡罗尔说。他指出，一些科学家担心这个想法定义得不够好，但他觉得，鉴于我们对量子引力所知，它是一个“自然的起点”。

Cyclic Cosmos

循环宇宙

Paul Steinhardt, a physicist at Princeton University, has another idea about what happened before the universe as we know it began. It stands in opposition to an idea that he helped to shape: the theory of inflation, which holds that after the big bang, spacetime expanded very quickly for a very short period. The inflation scenario is meant

to explain why the universe looks flat and similar in every place our telescopes can look.

普林斯顿大学物理学家保罗·斯坦哈特对已知宇宙开始之前发生了什么另有想法。它与他曾帮助塑造的一个想法相对立：暴胀理论，该理论认为大爆炸之后，时空在极短时期内极快膨胀。暴胀情景旨在解释为何宇宙看起来平坦，且在我们望远镜所能望见的每一处都相似。

After helping to establish inflation theory, however, Steinhardt started to doubt the idea, in part because it has required constant tweaking to keep it consistent with our measurements of the cosmos. "It's really hard to think of a historical example where that actually led to what turns out to be the right answer," Steinhardt says. "Almost always, that's a sign that the Titanic is sinking."

然而，在帮助确立暴胀理论之后，斯坦哈特开始怀疑这个想法，部分因为要保持它与我们对宇宙的测量一致，需要不断调整。“很难想到历史上有哪个例子，那种做法最终导向正确答案，”斯坦哈特说。“几乎总是，那是泰坦尼克号正在下沉的信号。”

Time to get in a lifeboat, he thought. So he came up with a cyclic universe: one that balloons significantly in size, as ours seems to be doing now, then shrinks a little and then starts expanding all over again. "When people think about contracting universes, they're usually thinking about things coming to a crunch," Steinhardt says—the cosmos collapsing back down into an infinitesimally small point. That's not what Steinhardt is talking about: he thinks the universe perhaps contracts slowly, to a smaller fraction of its size but not to nothing. That shrinking smooths things out in ways that inflation fails to explain, he says, while still producing a cosmos that

appears flat and the same in all directions.

该上救生艇了，他想。于是他提出了循环宇宙：一个像我们现在似乎正在做的那样显著胀大，然后稍微收缩，再重新开始膨胀的宇宙。“人们想到收缩宇宙时，通常想到的是走向大挤压，”斯坦哈特说——宇宙坍缩回无限小的一点。那不是斯坦哈特所说的：他认为宇宙或许缓慢收缩到其尺寸的一个较小分数，但不是到虚无。他说，那种收缩以暴胀无法解释的方式抚平事物，同时仍产生一个看起来平坦且各方向相同的宇宙。

Steinhardt adds that what looks like a big bang is actually not. The universe expands, then slowly contracts and then quickly goes back to expanding. The fast transition between contraction and expansion is not a bang but a "big bounce."

斯坦哈特补充说，看起来像大爆炸的其实不是。宇宙膨胀，然后缓慢收缩，再迅速回到膨胀。收缩与膨胀之间的快速过渡不是爆炸，而是“大反弹”。

Steinhardt hopes to test this idea not just by examining the past but also by taking data from the present and watching the future carefully. "It makes an obvious prediction, which is that the current phase of accelerated expansion can't continue forever," he says. "It must end." This idea, in turn, raises a new question: "Could it already be in the process of ending now?" he asks.

斯坦哈特希望不只通过考察过去，也通过采集当下数据并仔细观察未来来检验这一想法。“它做出一个明显的预言：当前加速膨胀阶段不可能永远继续，”他说。“它必须结束。”这一想法进而提出新问题：“它是否已经处在结束过程中？”他问。

Our measurements of the universe's expansion come from relatively faraway objects that emitted their light a long

time ago. Things could have changed, and we might not know yet, because the effects would be hard to measure. "We'd have to look at objects very close by in order to detect" a change in the universe's expansion, Steinhardt says. That's not cosmologists' forte, and they would have to develop new techniques and instruments to look nearby for such effects.

我们对宇宙膨胀的测量来自相对遥远的、很久以前发出其光的物体。事情可能已经改变，而我们或许还不知道，因为效应难以测量。“我们必须看非常近处的物体才能探测到”宇宙膨胀的变化，斯坦哈特说。那不是宇宙学家的强项，他们必须开发新技术与仪器来在近处寻找此类效应。

Even more intriguingly, Steinhardt says that because "nothing bad happens to space" during the contraction and bounce, information—even objects such as black holes—can pass from before the bounce to after. "There might be things in our observable universe that are from before," he says.

更引人入胜的是，斯坦哈特说，因为收缩与反弹期间“空间没有发生坏事”，信息——甚至黑洞这类对象——可以从反弹前传到反弹后。“我们可观测宇宙中或许有来自之前的东西，”他说。

Mirror Universe

镜像宇宙

Latham Boyle, a researcher at the Higgs Center for Theoretical Physics at the University of Edinburgh, who was formerly Steinhardt's graduate student, has another big

idea about the before before. Like the big bounce concept, Boyle's favored proposal is pretty simple conceptually, and it also avoids inflation. "There's the universe after the big bang and the universe before the big bang," he says, "and they're kind of mirror copies of each other."

爱丁堡大学希格斯理论物理中心研究员、曾是斯坦哈特研究生的莱瑟姆·博伊尔，对“之前的之前”另有大想法。像大反弹概念一样，博伊尔青睐的提案在概念上相当简单，也避开了暴胀。“有大爆炸之后的宇宙与大爆炸之前的宇宙，”他说，“它们有点像彼此的镜像副本。”

Picture this, Boyle says, like the points of two ice cream cones touching each other, with their contact representing the big bang. "Time marches away from the big bang in both directions," he says. On our side, it goes forward; on the mirror side, it goes backward. What happened before the big bang is the reflected opposite of what happened after. And that doesn't just include time: here there is matter; there there is antimatter. Here left is left; there left is right.

博伊尔说，把这想象成两个冰淇淋蛋筒的尖端彼此相触，接触点代表大爆炸。“时间从大爆炸向两个方向离去，”他说。在我们这边，它向前；在镜像那边，它向后。大爆炸之前发生的，是之后发生之事的反射对立。那不只包括时间：这里有物质，那里有反物质。这里左是左；那里左是右。

Boyle has ideas for observations that could support (or nullify) his theory, which is called the CPT-symmetric (charge-parity-time-symmetric) universe. For one, a CPT-symmetric universe wouldn't have sent gravitational waves shimmering through space from the beginning of the universe, as classical cosmology theories predict. Astronomers

have been hunting for such signals. If these waves are eventually detected, that would rule this idea out.

博伊尔对他称为 CPT 对称（电荷-宇称-时间对称）宇宙的理论有可支持（或否定）它的观测想法。其一，CPT 对称宇宙不会像经典宇宙学理论预言的那样，从宇宙开端就向空间发送闪烁的引力波。天文学家一直在搜寻此类信号。若这些波最终被探测到，就会排除这一想法。

Boyle's hypothesis also predicts that dark matter could be explained by a particular kind of neutrino. He hopes cosmological instruments will reveal more information about neutrinos soon. The model's connection to particle physics, among other aspects, makes this idea intriguing, Carroll says. "What I like here is the economy," Keating says, "and that it sticks its neck out," focusing on the kinds of specific, physical predictions experimentalists like him need.

博伊尔的假说还预言，暗物质可由一种特定的中微子解释。他希望宇宙学仪器不久会揭示更多关于中微子的信息。卡罗尔说，该模型与粒子物理的联系等方面使这一想法引人入胜。“我喜欢这里的经济性，”基廷说，“以及它把脖子伸出去，”聚焦于像他这样的实验家所需的那种具体物理预言。

Each of these scientists is attached to their own idea. But Lehnert isn't confident any of them will stand the test of time—whatever time is. "I think it's completely preposterous that, in the year 2025, we should understand the beginning of the universe," he says. In the grand scheme of things, in other words, we're still at the beginning of our journey of understanding. "Why not in the year 2,000,025 or whatever?"

这些科学家各执己见。但莱纳斯并不确信其中任何一个

能经受时间的考验——无论时间是什么。“我认为，在 2025 年我们就该理解宇宙的开端，完全是荒谬的，”他说。换言之，从大局看，我们仍处在理解之旅的开端。“为什么不是在 2,000,025 年或随便什么时候？”

And even if researchers think they are getting close, they could be approaching a false summit: that frustrating place that looks, when you're hiking, like the top of the mountain but is actually a mere bump blocking your view of the true peak. "In general, I think it's extremely plausible that there was something before the big bang," Carroll says, "but it's also very plausible that the big bang was truly the beginning. There's too much we're just unsure about, and I am a bit skeptical that the state of the art is good enough to allow us to draw any firm experimental or observational conclusions out of any of these models."

即便研究者们以为自己接近了，他们也可能正接近一座假山峰：那令人沮丧的地方，当你徒步时看起来像山顶，实际上只是挡住真正峰顶视线的一个小鼓包。“总的来说，我认为大爆炸之前有什么极为貌似合理，”卡罗尔说，“但大爆炸真正是开端也非常貌似合理。我们不确定的东西太多了，我有点怀疑当前技术水平是否足以让我们从这些模型中得出任何坚实的实验或观测结论。”

But cosmologists aren't studying our ultimate origins because they think the mystery will be resolved in their lifetime. Lehnert imagines himself as part of an intergenerational project helping humanity trek closer and closer to a truth we may never find. It's almost as if the mystery isn't within the realm of science. But science often involves probing things we can't access, at least at the start, Ismael says. Scientists predicted atoms before we could see them, and black holes and dark matter still lie

beyond our ability to detect them directly. "I think the benchmark for what counts as science has moved," she says. And it will continue to—including, perhaps, backward to the before that may not be a before.

但宇宙学家们研究我们的终极起源，并非因为他们认为谜题会在他们有生之年解决。莱纳斯把自己想象成一项跨代项目的一部分，帮助人类一步步趋近我们或许永远找不到的真理。这几乎仿佛谜题并不在科学领域之内。但科学常常涉及探查我们至少在开始时无法触及的事物，伊斯梅尔说。科学家们在能看见原子之前就预言了原子，黑洞与暗物质仍超出我们直接探测它们的能力。“我认为什么算作科学的基准已经移动了，”她说。它还会继续移动——包括，或许，向后移向那或许并不是“之前”的之前。

Why we love unexplainable mysteries

我们为何热爱无解之谜

Why we love unexplainable mysteries

By Clara Moskowitz · September 15, 2026 · edited by Clara Moskowitz

[https://www.scientificamerican.com/article/why-we-love-un](https://www.scientificamerican.com/article/why-we-love-unexplainable-mysteries)

The biggest questions in science inspire us even when we lack answers

科学中最大的问题即便缺乏答案也激励着我们

The most fascinating questions are often the hardest to answer. How did the universe begin? How did life evolve? What is the nature of consciousness? These questions inspired our special package on "Impossible Questions," which explores mysteries that are not just unsolved but potentially unsolvable.

最迷人的问题往往最难回答。宇宙如何开始？生命如何演化？意识的本质是什么？这些问题启发了我们关于“不可能的问题”的特别报道，探索那些不仅未解决、而且可能无法解决的谜题。

Such questions are also the driving force behind the Vox podcast *Unexplainable*, which delves into scientific unknowns large and small, from the nature of dark matter to

the science of human cells. The show, which debuted more than five years ago, also premiered as a video podcast on Netflix this summer. Over the years it has investigated mysteries such as whether plants can see, why humans cry, and what silence sounds like.

此类问题也是 Vox 播客《无法解释》(Unexplainable) 的驱动力；该节目深入科学未知，大到暗物质的本质，小到人类细胞的科学。节目五年多前首播，今年夏天也在 Netflix 上以视频播客形式首映。多年来它调查过植物能否看见、人类为何哭泣、沉默听起来像什么等谜题。

Scientific American spoke to *Unexplainable* host Noam Hassenfeld about dangerous unanswered questions, why bicycles are mysterious, and just how much we still don't know about the universe.

《科学美国人》采访了《无法解释》主持人诺姆·哈森菲尔德，谈危险的未解问题、自行车为何神秘，以及我们对宇宙仍有多少未知。

An edited transcript of the interview follows.

以下是经编辑的访谈实录。

What's the main idea behind the podcast?

播客背后的主要理念是什么？

Our show launched in early 2021, when the pandemic was going pretty intensely. There were lots of debates about how science works and how science should work. Something we really wanted to do was be honest that science is made by fallible scientists who have lives and make mistakes and are trying their best. And the most important thing is to be transparent about your mistakes and be open about what you don't know.

我们的节目在 2021 年初上线，当时疫情正激烈进行。关于科学如何运作、科学应当如何运作，有很多辩论。我们真正想做的是诚实：科学是由会犯错的科学家做出的，他们有生活，会犯错，并在尽力而为。最重要的是对你的错误保持透明，并对你不知道的事情保持开放。

There are some really big questions where we might not be close to answers. How big is the universe? How did life start? What's the point of it all? At the fundamental level, they are all trying to make sense of our existence.

有一些我们或许并不接近答案的真正大问题。宇宙有多大？生命如何开始？这一切的意义是什么？在根本层面上，它们都在试图理解我们的存在。

But I don't think you need to get to the huge cosmic existential questions to have really juicy, unanswered ones. I've found that our audience really appreciates finding out things that they don't know. Like, we don't know how bicycles work. This is one of my favorite ones.

但我不认为你需要走到巨大的宇宙存在问题，才能有真正有趣的未解问题。我发现我们的听众真正欣赏发现他们不知道的事情。比如，我们不知道自行车如何工作。这是我最喜欢的之一。

Wait, we don't understand how bicycles work?

等等，我们不理解自行车如何工作？

Yeah. This gets a little bit technical. It's a question of which physical theory actually explains how bicycles stay up. At a fundamental level, I think there's no very easy, simple way to explain why they do.

是的。这有点技术性。问题在于哪种物理理论实际解释自行车如何保持直立。在根本层面上，我认为没有非常容易、简单的方式来解释它们为何如此。

Wow! And that's a pretty different kind of unanswerable question than, for instance, how the universe began.

哇！而这与比如宇宙如何开始那种无法回答的问题，是相当不同类型的。

It's easy to assume that an unexplainable question is existential or cosmic. But we also do a lot of these really prosaic mysteries. So, why do we cry? Or should yogurt be its own state of matter? Or, you know, one of my favorites was an episode about the fact that some people can't burp. These basic questions are unexplainable just as much as the cosmic, where-did-we-come-from questions. Understanding exactly why these questions are unanswered lets us figure out exactly what we still need to figure out, and it also just helps me stop taking the everyday world for granted.

很容易假定无法解释的问题是存在性的或宇宙性的。但我们也做很多这些真正平凡的谜题。那么，我们为何哭泣？或者酸奶应否成为自己的物态？或者，你知道，我最喜欢的一集是关于有些人不会打嗝这一事实。这些基本问题与宇宙性的、我们从何处来的问题一样无法解释。准确理解这些问题为何未解，让我们弄清我们仍需弄清什么，也只是帮助我不再把日常世界视为理所当然。

*It's very easy to go throughout your day
and not confront that basic fact that al-
most everything everywhere is unknown.”*
—Noam Hassenfeld

“很容易过完一天，却不去面对那个基本事实：几乎一切、到处都是未知的。”——诺姆·哈森菲尔德

In the course of looking into these questions, have you come across a scientific mystery that seems to be not just currently unexplainable but unanswerable in theory?

在探究这些问题的过程中，你是否遇到过一个科学谜题，似乎不仅当前无法解释，而且在理论上也无法回答？

I want to say, "Never say never," because as I'm sure you know, many questions were once thought to be unsolvable, and then we figured them out, and we found ways to answer new questions we never even thought we would have.

我想说“永远别说永远”，因为我相信你知道，许多问题一度被认为无法解决，然后我们弄明白了，我们找到了回答我们从未想过会有新问题的方法。

But one that comes to mind is an episode we did on the question of whether the entire observable universe is inside a black hole. Is the universe just a black hole that is inside another universe that's inside a black hole, that's all black holes all the way down? At a certain level, that question is kind of unanswerable because you can't get outside the black hole. The scientist I talked to suggested building a particle accelerator that would go around the moon. So it's one of those things where maybe it's possible to answer it, if we did the most ridiculous thing imaginable.

但我想起的一个是我们做的一集，关于整个可观测宇宙是否在一个黑洞里面。宇宙是否只是一个黑洞，在另一个宇宙里面，而那个宇宙又在一个黑洞里面，一直是黑洞套黑洞？在某种层面上，那个问题有点无法回答，因为你无法走到黑洞外面。我交谈的科学家建议建造一座环绕月球的粒子加速器。所以那是那种或许有可能回答的事——如果我们做了最荒谬的可想象之事。

Are there some questions that should stay unexplained because trying to answer them would be too dangerous? The classic example might be the work that led to the atomic bomb. Should we have left those questions unanswered?

是否有些问题应当保持未解，因为试图回答它们会太危险？经典例子或许是通向原子弹的工作。我们本该让那些问题保持未答吗？

That's really tough. I think there are questions where finding the answer can be dangerous, like with the atomic bomb, although that also gave us carbon-free energy. And then there are also questions that maybe we can solve only if we don't do our research ethically. There are things that we've learned in the past through extremely unethical means. Someone once wrote into a Vox advice column asking, in the course of IVF, should I do everything I can to select the embryo that will have the least chance of disease? And then how far should I go? Should I select height and gender and eye color? You can see how it really spirals. I think having the impulse to solve things perfectly can both be impossible to meet and maybe lead you to a somewhat dangerous place.

那真的很难。我认为有些问题找到答案可能是危险的，比如原子弹，尽管那也给了我们无碳能源。然后还有一些问题，或许只有在我们不以伦理方式做研究时才能解决。过去有些我们通过极端不伦理手段学到的东西。有人曾写信给 Vox 建议专栏问：在体外受精过程中，我是否应尽一切可能选择患病几率最低的胚胎？然后我该走多远？我该选择身高、性别和眼睛颜色吗？你可以看到它如何真正螺旋上升。我认为拥有完美解决问题的冲动，既可能无法满足，也可能把你引向某种危险之地。

Have you ever felt surprised about just how much we don't know?

你是否曾对“我们不知道的有多少”感到惊讶？

I still remember an episode we did talking to researchers about the idea of whether there might have been a technological species before humans, like millions of years ago. We think we have this full picture of the past—we have all these fossils; we know what was going on. But in fact, we've barely found anything. You know, we've mapped less than one third of the ocean floor. We don't really know how smell works. Everything is just way more complicated than it seems. Ninety-five percent of the universe is made up of dark matter and dark energy, so we barely understand the vast majority of everything.

我仍记得我们与研究者交谈的一集，关于是否可能在人类之前存在过技术物种，比如数百万年前。我们以为自己完整的过去图景——我们有所有这些化石；我们知道发生了什么。但事实上，我们几乎什么也没找到。你知道，我们测绘了不到三分之一的海底。我们其实不知道嗅觉如何工作。一切都比看起来复杂得多。宇宙的百分之九十五由暗物质与暗能量构成，所以我们几乎不理解绝大多数一切。

You don't realize that what we know is like this little candle in the darkness. And when you're right next to it, it can feel like a lot. But then you zoom out, and you realize that we just don't really know that much. It's very easy to go throughout your day and not confront that basic fact that almost everything everywhere is unknown.

你意识不到我们所知道的就像黑暗中的这一小支蜡烛。当你紧挨着它时，它可能感觉很多。但然后你拉远，就会意识到我们其实知道得并不多。很容易过完一天，却

不去面对那个基本事实：几乎一切、到处都是未知的。

Does that ever feel discouraging?

那会让人气馁吗？

I don't know. On the one hand, it makes me feel that much more grateful for what we actually have discovered. How we've been able to slowly build this enormous body of knowledge in the face of all the unknowns. It definitely makes me not take that for granted. And I get excited about how much more there is to understand. I know that when I die, we are still going to understand basically nothing that's going on, but we are going to understand more than we do now. And there's a lot of room for everyone who comes afterward to figure more stuff out.

我不知道。一方面，它让我对我们实际已发现的东西更加感激。在所有未知面前，我们如何能够慢慢建立起这座知识的巨大身体。它确实让我不再把那视为理所当然。而我对还有多少可理解的感到兴奋。我知道当我死去时，我们仍将基本上什么也不理解正在发生的事，但我们会比现在理解得更多。而后来者有巨大空间去弄清更多东西。

When we started this show, we were like, are we going to run out of unexplainable things? And then very quickly we realized, definitely not. It kind of puts the magic back in the world a little bit. It makes you realize that this world is ridiculously complicated and that so many people are trying so hard to understand it. We've gotten so, so far, right? We have increased the mountain of knowledge to an absurd extent. And we still have so far to go.

当我们开始这个节目时，我们想：我们会不会用完无法解释的事物？然后很快我们意识到：绝对不会。它有点

把魔法放回世界一点点。它让你意识到这个世界荒谬地复杂，有那么多人如此努力地试图理解它。我们已经走了那么、那么远，对吧？我们把知识之山增加到了荒谬的程度。而我们仍有那么远的路要走。